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Stochastic Optimization of Trading Strategies in Sequential Electricity Markets

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Abstract

The increasing integration of renewable energy sources (RES) into electricity markets has heightened price and volume risks for market participants, due to the intermittent nature of wind and solar generation. Between the day-ahead market and real-time operations, significant discrepancies emerge between forecasts and actual realizations of key variables, such as renewable energy production and electricity prices. These uncertainties necessitate adaptive trading strategies that respond dynamically to evolving market conditions. This study explores optimal trading strategies for Vattenfall's management of the Hollandse Kust Zuid (HKZ) wind and solar park within the Dutch electricity market, focusing on the interaction between day-ahead and intraday trading. A two-stage stochastic optimization model is developed to account for production forecast errors, price volatility, and volume risks. In the first stage, bid ladders in the day-ahead market are optimized using scenario trees for price and production forecasts. In the second stage, deviations from day-ahead commitments are managed through intraday trading, leveraging updated market information and residual load adjustments. To quantify and manage risk, Conditional Value-at-Risk (CVaR) is incorporated into the objective function, enabling a structured trade-off between profit maximization and downside protection. Results show that while intraday trading increases expected profits by exploiting real-time opportunities, day-ahead participation serves as an effective hedging tool against uncertainty. The findings offer practical insights into how renewable energy producers can strategically navigate sequential electricity markets under uncertainty.

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1 Introduction

The integration of renewable energy sources (RES) has significantly increased uncertainty in electricity markets, particularly due to the variability of wind and solar generation.

As a result, market participants must make trading decisions across sequential markets, balancing profitability with risk management. The day-ahead (DA) market requires traders to commit to bids before knowing the actual production and prices, while the intraday (ID) market allows for adjustments based on updated forecasts. The increasing complexity of these sequential decisions, coupled with price volatility and negative prices, highlights the need for advanced optimization techniques to develop effective trading strategies, as discussed by Boomsma et al. (2014) and Möst and Keles (2010).

Existing approaches often rely on deterministic or simplistic stochastic models that fail to capture interdependencies between day-ahead (DA) and intraday (ID) markets. While prior research highlights price and volume risks, many studies overlook the conditional relationships between uncertainties. This study develops a two-stage stochastic optimization model integrating day-ahead bidding with intraday market adjustments, explicitly incorporating market uncertainties via a scenario-based approach. Unlike previous works, this study models the stochastic processes of day-ahead and intraday electricity prices, as well as their interdependencies within and across both markets, while utilizing Vattenfall's proprietary generation forecasts.¹ Simulations capture price uncertainty and its role in stochastic optimization. The study focuses on the Dutch electricity market,² which shares design features with liberalized markets across Europe. The proposed model optimizes day-ahead bid ladders while dynamically adjusting intraday commitments, incorporating Conditional Value-at-Risk (CVaR) for risk management, following the approach introduced by Conejo et al. (2010).

The findings of this study offer insights for renewable energy traders by evaluating different risk preferences and their impact on trading performance. While risk-neutral strategies aim to maximize expected profits, risk-averse approaches prioritize stability and minimize exposure to unfavorable market conditions. By constructing a scenario tree that reflects price and production uncertainties, this study provides a robust

¹ Vattenfall's internal forecasting model provides production estimates, reducing the need for additional generation uncertainty modeling. Source: Vattenfall.

² The Dutch intraday market operates with both continuous trading (XBID) and periodic auctions, increasing complexity compared to fully continuous markets like Germany.

framework for improving decision-making in sequential electricity markets.

The remainder of this thesis is structured as follows. Section 2 provides an overview of the European electricity market framework, focusing on the structure and mechanisms of the day-ahead, intraday, and balancing markets. Section 3 reviews existing research on stochastic optimization in electricity markets, highlighting key gaps and methodologies. Section 4 examines the statistical properties of electricity prices, including volatility, mean reversion, and residual load dynamics. Section 5 details the methodological approach, covering forecasting techniques, scenario generation, and the construction of a structured scenario tree to capture market uncertainties, which serves as an input for the optimization model introduced in Section 6. This two-stage stochastic optimization framework is designed to optimize bid strategies across the day-ahead and intraday markets while managing risk exposure. Section 7 evaluates different trading strategies by analyzing their profitability and risk implications. Finally, Section 8 concludes the study with key findings and suggests future research directions.

2 European Electricity Market Framework

The European electricity market consists of three primary segments: the day-ahead market (DAM), the intraday market (IDM), and the balancing reserve market. These markets facilitate trading and system stability across different time horizons. While the DAM and IDM operate in a harmonized framework across Europe, balancing reserves remain organized on a national level with distinct mechanisms for frequency control. Rising shares of renewable energy sources (RES) have increased the need for near real-time decision-making, particularly in the interaction between day-ahead and intraday markets.

Figure 1 provides an overview of the sequential market structure. The DAM is a centralized auction where participants submit bids by 12:00 CET for delivery the following day, with a uniform pricing mechanism determining the market-clearing price. This market sets the foundation for energy commitments, but given uncertainties in RES production, market participants often turn to the IDM for adjustments. The IDM opens at 15:00 CET on the previous day, offering continuous trading

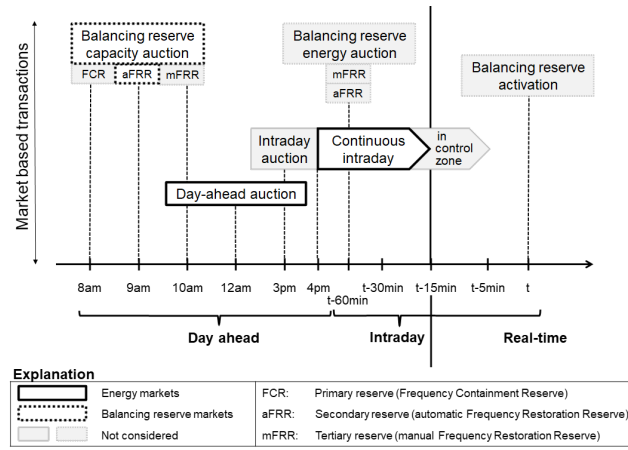


Figure 1: Timeline of market-based transactions in the Dutch electricity market.

opportunities in quarter-hourly products up to 5 minutes before delivery.³⁴

To model intraday market conditions effectively, the ID3 price index is commonly used, representing the volume-weighted average of trades executed in a three-hour window before delivery. For example, the ID3 price for 12:00–13:00 is derived from transactions between 9:00 and 11:30, capturing price signals during the final stages of intraday trading when market participants actively balance their positions, as described by Ottesen, Tomasgard, and Fleten (2018) and Laur et al. (2020). We assume that the trader operates as a price-taker, meaning they can always find a buyer for their electricity at the ID3 price. While this assumption may have been unrealistic in the early years of intraday trading due to low market liquidity, the situation has significantly improved over time. Today, intraday markets are liquid enough to justify this approach, especially given the relatively small trading volumes in the portfolio under consideration. Since the focus of this paper is on optimizing day-ahead trading strategies, this assumption is deemed reasonable.

Balancing reserve markets provide reserves that can be activated to maintain grid stability. The automatic Frequency Restoration Reserve (aFRR) market offers rapid, automatic response to frequency deviations, while the manual Frequency Restoration Reserve (mFRR) requires operator intervention to manage larger imbalances. The

³ While shorter time resolution products (e.g., 15-minute contracts) are available in the Dutch IDM and provide more granular price signals, this study focuses on ID3 hourly prices as a representative indicator of intraday market conditions. Including additional product resolutions would primarily result in substitution effects rather than fundamentally altering trading strategies. Therefore, we optimize trading strategies across decision stages using ID3 hourly prices as the primary benchmark for the intraday market.

⁴ It is worth noting that the market design has recently changed. As of early 2025, the intraday market opens at 14:00 CET and trades continue up to one second before delivery. Additionally, the day-ahead market is expected to transition to quarter-hourly products soon. The design presented here reflects the market conditions at the beginning of this thesis.

procurement of aFRR in the Netherlands is organized in a two-stage procedure. The first stage, the so-called capacity market auction for the day ahead, takes place between 07:00 and 10:00 CET on the day before delivery and determines the reserve providers for the following day. In this auction, prequalified reserve providers submit bids specifying a capacity price (in EUR/MW) and a volume (in MW). Providers may submit multiple distinct bids. Unlike some other markets that use fixed time blocks, the Dutch aFRR market operates with hourly time slices, allowing for more granular and flexible participation. Separate auctions are held for positive (upward regulation) and negative (downward regulation) reserves, and the remuneration follows a marginal pricing (pay-as-cleared) mechanism, meaning all accepted bids receive the same price, equal to the highest accepted bid. The second stage, the aFRR energy market auction, takes place 45 minutes before the intraday market gate closure and determines activation prices and the merit order for reserve activation. The Netherlands is also integrated into the European PICASSO platform, which aims to harmonize aFRR markets across participating countries and has contributed to reducing imbalance price volatility.

This study focuses solely on DAM and IDM trading dynamics, omitting balancing markets to maintain tractability. For simplicity, balancing reserve markets are excluded, allowing a concentrated focus on DAM and IDM interactions. By optimizing bidding strategies and managing intraday deviations, this study offers actionable insights into mitigating the challenges posed by renewable energy integration. The findings contribute to the broader objective of enhancing market efficiency, improving risk management, and facilitating a smoother transition toward sustainable energy systems.

	Deterministic / realized	Stochastic
Stage 1	<i>deterministic:</i> Expected value RES generation forecast, technical and market constraints	<i>scenarios i:</i> Day-ahead RES generation forecast, day-ahead market prices
Stage 2	<i>scenarios i:</i> Day-ahead market result and commitment and RES generation forecast	<i>scenarios j:</i> Intraday update RES generation forecast, intraday market prices

Table 1: Overview over information available to the optimization as deterministic or realized information and as stochastic information in form of scenarios.

In conclusion, the trader must navigate two key markets: the day-ahead market, which consists of 24 hourly products, and the intraday market, represented by 24 hourly ID3 prices in this paper to simplify complexity. Given the uncertainty in

prices across these markets and the variability in renewable generation, the trader faces a highly complex problem involving numerous decision variables.

All data, sources, and timeframes are listed (see Table 7 for reference).

3 Literature Review

This chapter reviews the literature on stochastic optimization techniques for electricity market trading, emphasizing multi-stage optimization frameworks that incorporate market interactions and associated uncertainties.

Electricity market participants face two primary short-term optimization challenges: the unit commitment problem (optimal dispatch) and the optimal trading problem. The unit commitment problem involves delivering a predefined energy schedule or providing flexibility at minimal cost, while the optimal trading problem extends this by incorporating market expectations to optimize bids and maximize profits. Since market commitments are uncertain *ex ante*, unit commitment is often modeled implicitly rather than explicitly.

Deterministic approaches, typically formulated as Mixed-Integer Linear Programming (MILP) models, have been widely used to handle technical constraints, as discussed by Birge and Louveaux (2011). However, deterministic models fail to account for the increasing uncertainties associated with renewable generation and volatile market prices. Consequently, stochastic programming has emerged as a preferred framework. Zheng, Wang, and Liu (2015) categorize stochastic approaches into stochastic programming, robust optimization, and stochastic dynamic programming, each addressing uncertainty in different ways.

Early works by Swider (2007) and Boomsma et al. (2014) emphasized the necessity of incorporating stochasticity into market modeling, particularly for balancing-power and day-ahead markets. Swider (2007) developed foundational stochastic models for bidding in balancing-power markets, while Boomsma et al. (2014) extended these ideas to multi-stage stochastic programs, capturing coordinated bidding across spot and balancing-power markets for the Nordic case. Building on these foundations, Fleten and Kristoffersen (2007) applied stochastic programming to hydropower bidding strategies, balancing risk and profitability under uncertain market conditions. Similarly, Ottesen, Tomasgard, and Fleten (2018) explored demand-side management portfolios across multiple market segments, showcasing the flexibility offered by

multi-stage stochastic programming in addressing uncertainties across temporal scales.

Deng and Oren (2006) advanced the modeling of sequential markets by integrating day-ahead, balancing-power, and continuous intraday markets, incorporating price correlations among market segments. However, their model lacked explicit consideration of balancing-power requests and continuous trading dynamics, leaving gaps that this thesis seeks to address. Recent studies have increasingly focused on integrating day-ahead (DA) and intraday (ID) markets. Kiesel and Paraschiv (2017) and Kremer, Kiesel, and Paraschiv (2021) highlighted the significance of price volatility in intraday markets driven by renewable forecast updates. These studies underscored the importance of strategic order placement and timing in continuous trading environments.

Ottesen, Tomasgard, and Fleten (2018) and Boomsma et al. (2014) modeled interactions between day-ahead, intraday, and balancing-power markets, emphasizing sequential decision-making in European electricity markets. However, their assumptions of single-stage intraday trading and limited modeling of balancing-power remuneration indicate a need for further exploration of comprehensive multi-stage frameworks.

Risk metrics have become an integral part of electricity market optimization models. Conejo, Carrión, and Morales (2010) introduced Conditional Value-at-Risk (CVaR) as a measure of risk aversion in electricity market bidding. Laur et al. (2020) extended this concept to local energy markets, considering risk implications across multiple market segments. In the context of retailers and renewable generators, Russo and Bertsch (2020) examined the impact of distributed solar PV generation on short-term market risks, highlighting the challenges of volume risk management in intraday markets. Wozabal and Rameseder (2020) proposed a two-stage stochastic model for day-ahead and intraday market trading, addressing biases introduced by simplistic deterministic approaches. Their work provided a robust framework for understanding market dynamics under uncertainty by modeling the interdependencies between load, prices, and renewable generation.

Russo et al. (2021) investigated retailer decision-making under increasing decentralized solar generation. They proposed a two-stage stochastic optimization framework to jointly model interdependencies between day-ahead and intraday markets, incorporating uncertainty from rooftop solar PV generation. Their findings emphasized the importance of real-time adjustments to mitigate volume and price risks and demonstrated the value of combining econometric modeling with stochastic optimization for short-term

electricity markets.

While significant advancements have been made in stochastic optimization for electricity markets, existing models often fail to fully capture the sequential nature of day-ahead and intraday markets. Key gaps include limited consideration of continuous intraday trading dynamics, incomplete modeling of balancing power requests and remunerations, and insufficient exploration of the interplay between price and volume risks across market segments.

This thesis addresses these gaps by developing a two-stage stochastic optimization framework for the Dutch electricity market, which is known for its complexity, high degree of uncertainty, and often inexplicable price patterns—likely driven by high market concentration. The proposed model optimizes bid ladders in the day-ahead market and manages deviations in the intraday market, incorporating dynamic market conditions and risk metrics such as Conditional Value-at-Risk (CVaR).

By extending existing methodologies to include comprehensive market dynamics and uncertainty, this research contributes to the development of robust trading strategies for renewable energy portfolios.

4 Stylized Facts of Dutch Electricity Prices

Electricity markets exhibit pronounced volatility and distinct statistical characteristics that have direct implications for trading strategies and risk management. This chapter investigates the key stylized facts of day-ahead and intraday electricity prices, as well as residual load dynamics, with a focus on statistical properties, volatility patterns, and their operational significance. The analysis is based on hourly market data from **January 2023 to October 2024**, capturing recent developments and enabling the identification of seasonal and structural price behaviors.

The time series of day-ahead electricity prices over the studied period, depicted in Figure 2, exhibits substantial variability. Prices demonstrate pronounced positive and negative spikes, reflecting the volatility of electricity markets driven by fluctuating supply-demand balances, renewable energy variability, and external factors such as weather events. The histogram in Figure 3 further reveals the price distribution, with a concentration of values around the mean and fat tails indicating occasional extreme price deviations. This distributional pattern underscores the challenges faced by market participants in forecasting and risk management.

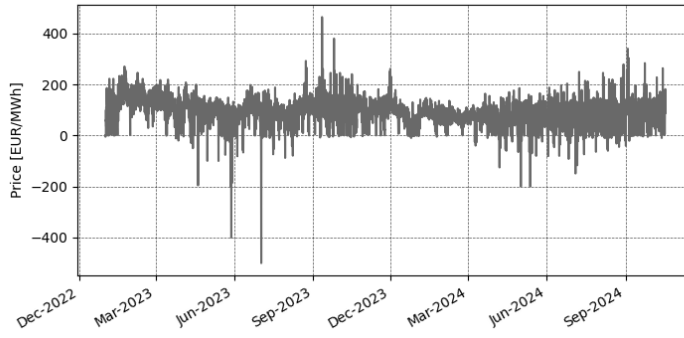


Figure 2: Hourly day-ahead electricity prices (EUR/MWh).

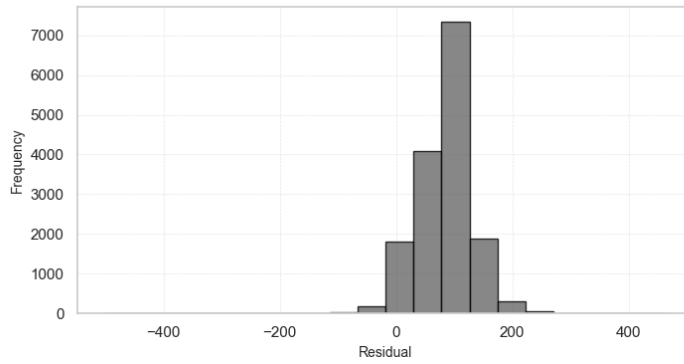


Figure 3: Histogram of day-ahead electricity prices.

The descriptive statistics of day-ahead prices, summarized in Table 2, provide additional insights. The mean price is 83.96 €/MWh, with a standard deviation of 47.78 €/MWh, highlighting the variability in market conditions. The distribution is slightly negatively skewed (-0.77), indicating a tendency for more frequent low-price occurrences, and the high kurtosis (8.07) confirms the presence of extreme price events. These characteristics underline the importance of robust statistical models to capture these non-standard features to deal with non-normality of data.

Table 2: Descriptive statistics of day-ahead electricity prices.

Metric	Mean	Std Dev	Min	Max	Skewness	Kurtosis
Unit	€/MWh	€/MWh	€/MWh	€/MWh	-	-
DA Prices	83.96	47.78	-500	463.77	-0.77	8.07

The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots for day-ahead electricity prices, shown in Figure 4, provide valuable insights into the temporal dependencies. The ACF plot reveals strong periodic patterns, with significant correlations at weekly intervals, aligning with operational cycles in electricity demand and generation. This periodicity highlights the structured nature

of energy consumption and production schedules.

The PACF plot highlights the lag structure most relevant for predicting future prices. The initial significant lags indicate that recent prices heavily influence short-term price movements, making autoregressive models suitable for capturing these dynamics. However, the rapid decay of partial autocorrelations suggests diminishing influence from prices beyond the immediate past. These findings emphasize the importance of incorporating lagged price terms and periodic patterns into predictive models for day-ahead electricity prices.

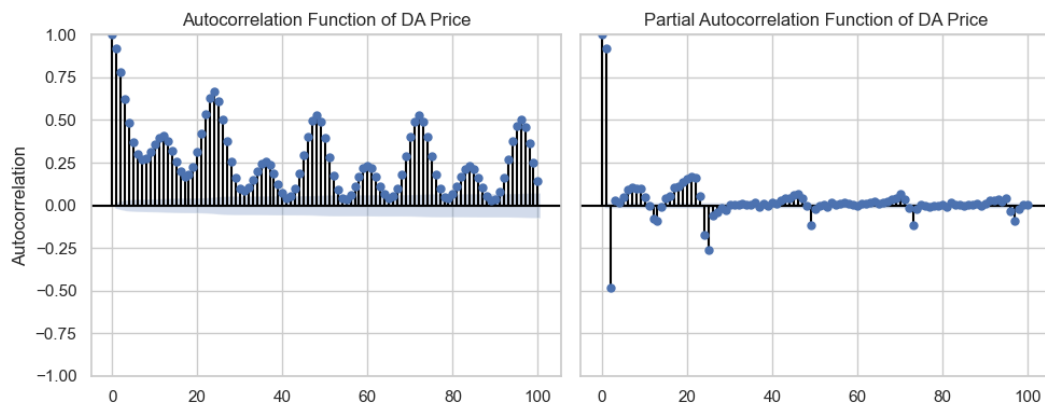


Figure 4: Autocorrelation function (ACF) and partial autocorrelation function (PACF) of day-ahead electricity prices.

Residual load (RL), defined as the difference between forecasted demand and renewable energy generation, plays a critical role in electricity markets. The time series of residual load, shown in Figure 5, reveals distinct seasonal patterns, with high values corresponding to periods of low renewable generation and increased reliance on conventional energy sources. Autocorrelation (ACF) and partial autocorrelation (PACF) plots in Figure 6 demonstrate strong weekly periodicity, aligning with the operational cycles of electricity demand and generation.

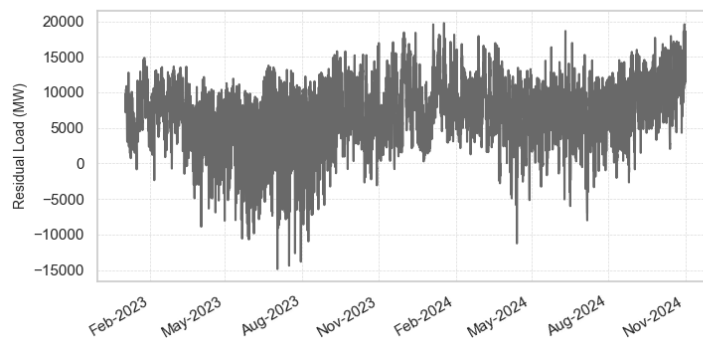


Figure 5: Time series of hourly residual load (MW).

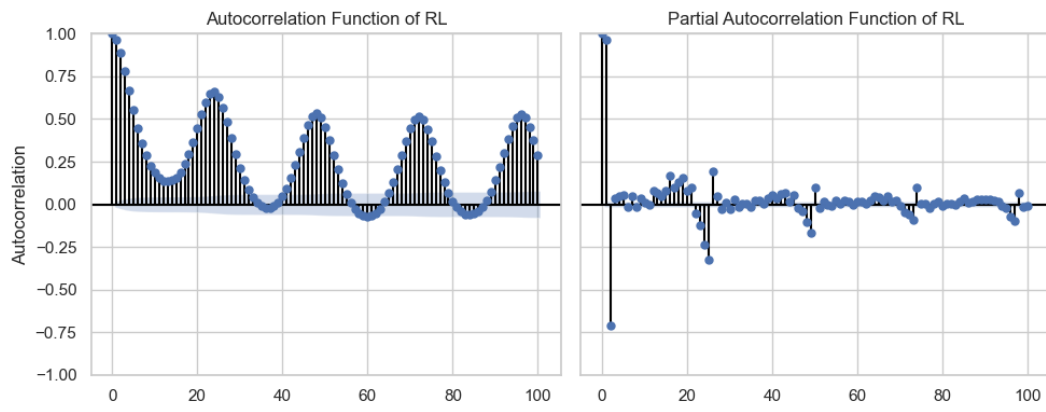


Figure 6: Autocorrelation and partial autocorrelation of residual load.

The descriptive statistics in Table 3 reveal a high mean of 6912 MW and substantial variability, indicating the dynamic interaction between demand and renewable supply. The distribution exhibits a slight left skew with fewer extreme outliers than a normal distribution, making it relatively symmetrical with a flatter peak.

Table 3: Descriptive statistics of residual load.

Metric	Mean	Std Dev	Min	Max	Skewness	Kurtosis
Unit	MW	MW	MW	MW	-	-
RL	6912	4507	-14878	19750	-0.713	1.206

Electricity prices also exhibit clustering in volatility, as illustrated in Figure 7. The chart displays the rolling 30-period standard deviation of day-ahead prices, capturing short-term fluctuations in price variability. Periods of elevated volatility often align with disruptions in renewable generation or sudden shifts in demand, highlighting the sensitivity of electricity markets to external shocks. This clustering behavior is a well-documented stylized fact in electricity markets, as noted by Cont (2001), stemming from their dependency on weather-driven renewable energy sources and the need for real-time supply-demand balancing.

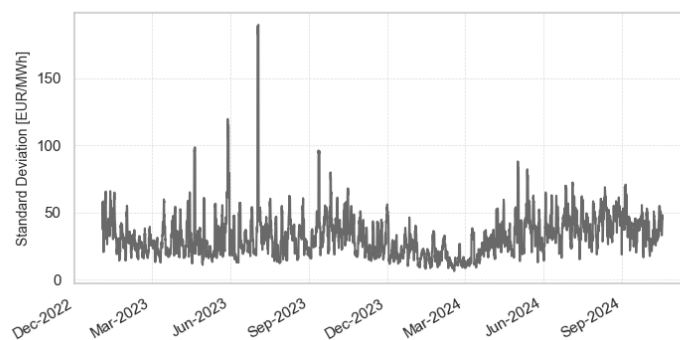


Figure 7: Volatility clustering in day-ahead electricity prices, measured as the rolling 30-period standard deviation (EUR/MWh).

A comparison between day-ahead and intraday (ID3) prices reveals significant differences in their statistical characteristics. As illustrated in Figure 8, the ID3 price time series exhibits a higher average price of 85.83 €/MWh and greater variability, with a standard deviation of 55.80 €/MWh, compared to day-ahead prices. These distinctions are summarized in Table 4. The increased volatility and elevated price levels in the intraday market can be attributed to real-time adjustments required to balance deviations from day-ahead commitments, primarily driven by updated forecasts of renewable generation and electricity demand.

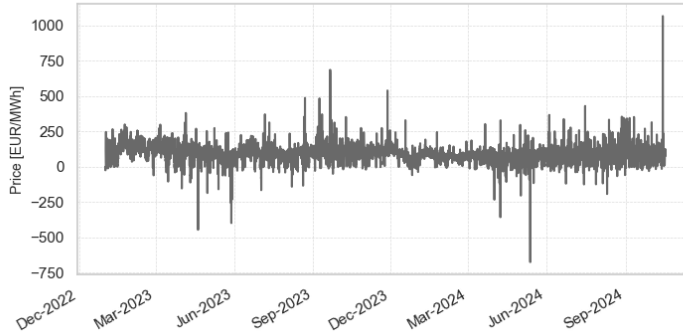


Figure 8: Hourly intraday ID3 electricity prices (EUR/MWh).

Table 4: Descriptive statistics of intraday ID3 electricity prices.

Metric	Mean	Std Dev	Min	Max	Skewness	Kurtosis
Unit	€/MWh	€/MWh	€/MWh	€/MWh	-	-
ID3 Prices	85.83	55.80	-675.83	1064.80	0.515	22.381

Further statistical features of ID3 prices are visualized in Figures 30 to 34. These include the distribution of prices (histogram), autocorrelation and partial autocorrelation functions (ACF and PACF), volatility clustering, and signs of mean reversion. For comparison, similar analyses for day-ahead prices are provided above, enabling a side-by-side evaluation of both market segments.

Electricity prices exhibit extreme volatility, heavy tails, and negative values due to imbalances between renewable generation and demand. To address these challenges, the Area Hyperbolic Sine (Arsinh) transformation is applied to stabilize variance while preserving data structure. Unlike the logarithmic transformation, which requires strictly positive inputs, the Arsinh transformation is defined for all real numbers, making it well-suited for electricity markets. Mathematically, it is expressed as:

$$y = \sinh^{-1}(x) = \ln \left(x + \sqrt{x^2 + 1} \right), \quad (1)$$

where x represents the original price data, and y is the transformed value. The Arsinh transformation effectively compresses extreme values, stabilizing variance while preserving the symmetry of the distribution. As demonstrated by Uniejewski, Weron, and Ziel (2018), applying variance-stabilizing transformations such as Arsinh significantly improves electricity price forecasting by reducing the influence of extreme spikes and accommodating negative prices.

Figure 9 compares original and Arsinh-transformed day-ahead prices, highlighting its ability to reduce variability. The mean-reversion property of day-ahead prices, depicted in Figure 10, further supports the existence of a long-term equilibrium, crucial for forecasting models.

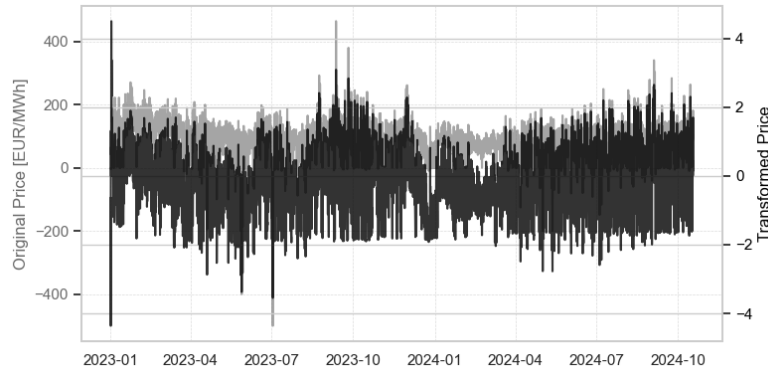


Figure 9: Original and arcsinh-Transformed day-ahead electricity prices.

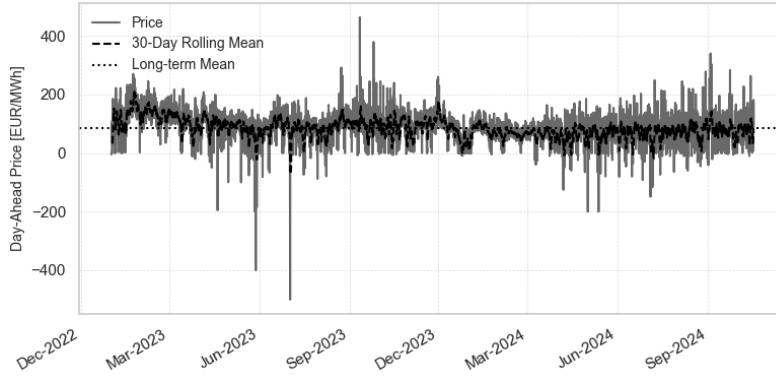


Figure 10: Mean reversion in day-ahead electricity prices.

In summary, the Arsinh transformation is a robust method for handling electricity price fluctuations, stabilizing variance, and improving statistical modeling. Its ability to process negative values and extreme spikes makes it particularly valuable for forecasting and trading strategies in electricity markets, as shown by Uniejewski, Weron, and Ziel (2018).

5 Methodology

Before modeling uncertainties in the day-ahead and intraday markets, a comprehensive data preprocessing step is essential to ensure statistical reliability. Electricity price and residual load time series often contain anomalies such as negative values, extreme outliers, and missing observations, all of which can significantly distort downstream scenario generation and inference.

To address these issues, the data is first inspected and treated to ensure consistency across all time periods. Missing values are addressed through linear interpolation, which estimates the missing data points based on a straight-line fit between neighboring observed values. This method ensures continuity in the time series without introducing abrupt structural changes, making it suitable for high-frequency electricity market data. Outlier handling is then performed by capping values beyond the 2.5% and 97.5% percentiles within each market segment. These extreme values are replaced with their corresponding threshold values, following the approach of Janczura et al. (2013). This method improves the robustness of the dataset while preserving its underlying statistical characteristics, thus providing a stable foundation for subsequent modeling and scenario generation tasks.

In the following sections, we describe the process of generating and analyzing price scenarios based on this methodology. The framework is then applied to a one-month case study covering the period from **September 17, 2024**, to **October 16, 2024**, to illustrate the model's performance and practical applicability. For each day within this sample, the model is retrained using all historical data available up to that point in time (t) to generate a deterministic day-ahead forecast. This daily iteration enables us to produce a rolling sequence of forecasts, construct corresponding scenario trees, reduce them via clustering techniques, and optimize bid strategies accordingly. This dynamic approach ensures realistic simulation of day-to-day market decision-making under uncertainty.

5.1 Modeling Uncertainty in the Day-Ahead Market

This study utilizes a multiple linear regression framework to model uncertainties in day-ahead electricity markets. By leveraging historical data (see Table 7) and carefully engineered features, the model captures dynamic relationships between

key variables, such as forecasted residual load, lagged prices, and hourly market behaviors. The methodology focuses on developing a robust statistical approach for forecasting day-ahead prices while addressing potential limitations and ensuring interpretability of results.

The choice of this regression model is motivated by its ability to provide a parsimonious representation of the complex dynamics inherent in electricity prices while maintaining interpretability. Electricity prices exhibit strong autocorrelation and periodic patterns, as evidenced by the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots presented in Figure 4. The ACF highlights significant serial correlation, particularly at daily (24-hour) and weekly (168-hour) lags, while the PACF confirms the presence of short-term dependencies and weekly seasonality. These observations validate the inclusion of lagged prices at 24, 48, and 168-hour intervals to account for temporal dependencies and cyclical patterns in the data.

Several alternative modeling approaches, including more complex machine learning methods, were explored during the development process. However, the multiple linear regression model was chosen due to its parsimony and effectiveness in capturing the variability in electricity prices while avoiding overfitting. This approach balances complexity and interpretability, ensuring that the model remains both computationally efficient and practically applicable for forecasting purposes.

The regression model incorporates several predictive variables, including a weekend indicator, rolling price statistics, lagged prices, modular price cycles, and residual load lags. These features account for structural differences in demand, price volatility, and temporal dependencies. For instance, the weekend indicator differentiates between weekday and weekend demand patterns, while rolling minimum and maximum prices reflect daily price dynamics. Lagged prices at 24, 48, and 168-hour intervals capture autocorrelation, which is a well-known characteristic of electricity prices. Similarly, residual load lags encapsulate the effect of load imbalances on price formation.

The residual load used in the regression model is forecasted using the Vattenfall forecasting model, ensuring reliable predictions for the upcoming day. Importantly, we use the day-ahead forecasts generated 8 hours prior to avoid any lookahead bias. This allows us to reference the same time period (t) for residual load forecasts in the model, as they are forecasts available in advance of the trading day. By

incorporating forecasted residual load values, the model captures the critical relationship between expected load imbalances and price formation without compromising its predictive integrity.

To further understand the relationships between predictive variables, Figure 11 shows the correlation matrix of the variables used in the regression model. The correlation matrix highlights strong positive correlations among lagged prices **DA_price_lag_24**, **DA_price_lag_48**, and **DA_price_lag_168**, reflecting their shared contribution to capturing price trends and autocorrelation. Residual load forecast lags **rl_lag_0**, **rl_lag_24**, etc. also exhibit meaningful correlations with day-ahead prices, emphasizing the influence of load imbalances on market dynamics. The modular price cycle **P_DA_modular** shows moderate correlations with lagged prices, capturing cyclical variations in price patterns. These correlations justify the inclusion of these variables in the regression model, as they collectively contribute to explaining the variability in day-ahead prices.

By combining insights from ACF and PACF analysis with carefully selected features, the proposed regression model effectively captures the core drivers of day-ahead electricity price variability. This parsimonious approach not only ensures robust forecasting performance but also provides interpretability, making it a practical choice for analyzing and understanding the dynamics of electricity markets.

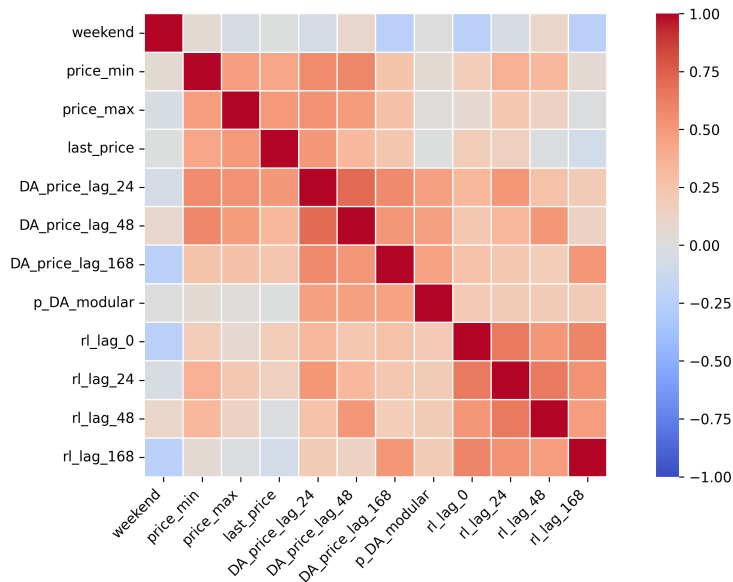


Figure 11: Correlation matrix of variables used in the regression model.

The multiple linear regression equation for the day-ahead market is given as follows:

$$\begin{aligned}
DA\ Price_t = & \beta_0 + \beta_1 \textit{Weekend}_t + \beta_2 \textit{Price Min}_{t-24} + \beta_3 \textit{Price Max}_{t-24} + \beta_4 \textit{Last Price}_{t-1} \\
& + \sum_{k \in \{24, 48, 168\}} \beta_{k+4} DA\ Price\ Lag_k + \sum_{k \in \{0, 24, 48, 168\}} \beta_{k+7} RL\ Lag_k + \epsilon_t
\end{aligned} \tag{2}$$

where:

- ***DA Price_t***: Day-ahead market clearing price at time t .
- ***Weekend_t***: Binary variable indicating weekends (1 for weekends, 0 for weekdays).
- ***Price Min_{t-24}***: Rolling minimum day-ahead price over the past 24 hours.
- ***Price Max_{t-24}***: Rolling maximum day-ahead price over the past 24 hours.
- ***Last Price_{t-1}***: Day-ahead price at 23:00 on the previous day.
- ***DA Price Lag_k***: Lagged day-ahead prices at 24, 48, and 168-hour intervals.
- ***RL Lag_k***: Lagged residual load forecast values at 0, 24, 48, and 168-hour intervals.
- ϵ_t : Error term capturing unexplained variability.

5

5.1.1 Regression Results and Interpretation

The regression results, summarized in Table 8, demonstrate that the model explains a substantial proportion of the variance in day-ahead electricity prices, with an R^2 value of 0.784. This indicates strong predictive capability, suggesting that historical price patterns and fundamental market drivers are well captured within the model. The high adjusted R^2 confirms that the inclusion of additional explanatory variables does not introduce unnecessary complexity but rather enhances explanatory power.

The coefficients of lagged prices at 24, 48, and 168 hours are all statistically significant and exhibit substantial magnitude, reinforcing the well-documented autocorrelation in electricity prices. This persistence in price movements aligns with expectations in energy markets, where past price information is a key determinant of future price formation. Furthermore, the coefficient of the last observed price (P_{t-1}^{Last}) is highly significant and positive, reflecting the immediate impact of recent price levels on the next period's forecast.

The weekend indicator variable (W_t) is negative and significant, indicating a systematic decline in day-ahead prices on weekends. This result is consistent with lower

electricity demand and reduced market activity during weekends, leading to price reductions. The minimum and maximum price variables (P_{t-24}^{Min} and P_{t-24}^{Max}) also have negative coefficients, suggesting that extreme price fluctuations in the previous day contribute to moderating effects in the current period.

Residual load lags play a crucial role in price determination, with mixed directional effects. The contemporaneous residual load (RL_t) has a small but statistically significant positive coefficient, indicating that higher net load (demand minus renewable generation) leads to upward pressure on prices. However, the negative coefficients of lagged residual load terms (RL_{t-24} , RL_{t-48} , and RL_{t-168}) suggest that price adjustments occur over time as the market corrects for imbalances. The inverse relationship between past residual load levels and present prices highlights the dynamic balancing mechanisms in electricity markets.

In conclusion, the regression results affirm the significance of lagged prices, residual load dynamics, and market fundamentals in shaping day-ahead electricity prices. The model captures key market dependencies, and its coefficients remain stable over time, reinforcing its reliability for forecasting. However, residual diagnostics (discussed in the next section) indicate potential improvements, particularly in addressing heteroscedasticity and residual autocorrelation, which could further enhance forecasting accuracy.

5.1.2 Residual Analysis and Model Diagnostics

A comprehensive residual analysis was conducted to evaluate the validity of the Ordinary Least Squares (OLS) assumptions, including normality, homoscedasticity, and the absence of autocorrelation. These assumptions are critical for ensuring unbiased and efficient parameter estimates in the day-ahead electricity price forecasting model.

The normality of residuals was assessed using a Quantile-Quantile (QQ) plot (Figure 12b) and a histogram of residuals (Figure 12a). The QQ plot shows deviations from the theoretical normal distribution, particularly at the tails, indicating potential heavy-tailed behavior in the residuals. This suggests that extreme price movements occur more frequently than expected under a normal distribution, which is a common feature in electricity markets. The histogram of residuals, while approximately symmetric around zero, further confirms excess kurtosis, implying that the residuals exhibit

more extreme values than a Gaussian distribution would predict.

Further investigation into the residuals was conducted using the residuals vs. fitted values plot (Figure 12c). Ideally, residuals should be randomly scattered around zero, indicating that the model captures all systematic relationships in the data. However, the plot exhibits a distinct funnel-shaped pattern, suggesting heteroscedasticity, where the variance of residuals changes with the magnitude of fitted values. This is a common phenomenon in electricity price data, where extreme price fluctuations often occur under specific market conditions, such as periods of low demand or high renewable generation.

A key economic explanation for this heteroscedasticity lies in the non-linear structure of the merit order curve used in electricity price formation. At very high and very low prices, the number of active market participants becomes limited, meaning that even small changes in demand or supply can result in disproportionately large price shifts. Consequently, price volatility is directly linked to the prevailing price level, with higher volatility observed at the extremes of the distribution.

Additionally, the sharp diagonal boundaries in the plot correspond to cases where the fitted value plus residual equals either 0 €/MWh or 175 €/MWh, indicating censoring effects at the limits of the observed price range. These patterns reflect real-world constraints—namely, the non-negativity of electricity prices and an upper cap in the dataset—both of which influence the residual distribution near the extremes.

To further examine the residuals' time dependence, Figure 12d presents the residuals over time. A well-calibrated model should produce residuals that resemble white noise, meaning they exhibit no systematic pattern or autocorrelation. However, this plot indicates periods of residual clustering, where positive and negative residuals persist over time, suggesting autocorrelation. This violates the assumption of independent errors and implies that past residuals contain information relevant for future predictions. The presence of autocorrelation suggests that the model may benefit from additional lag structures or more advanced econometric techniques, such as autoregressive models or time-varying volatility frameworks.

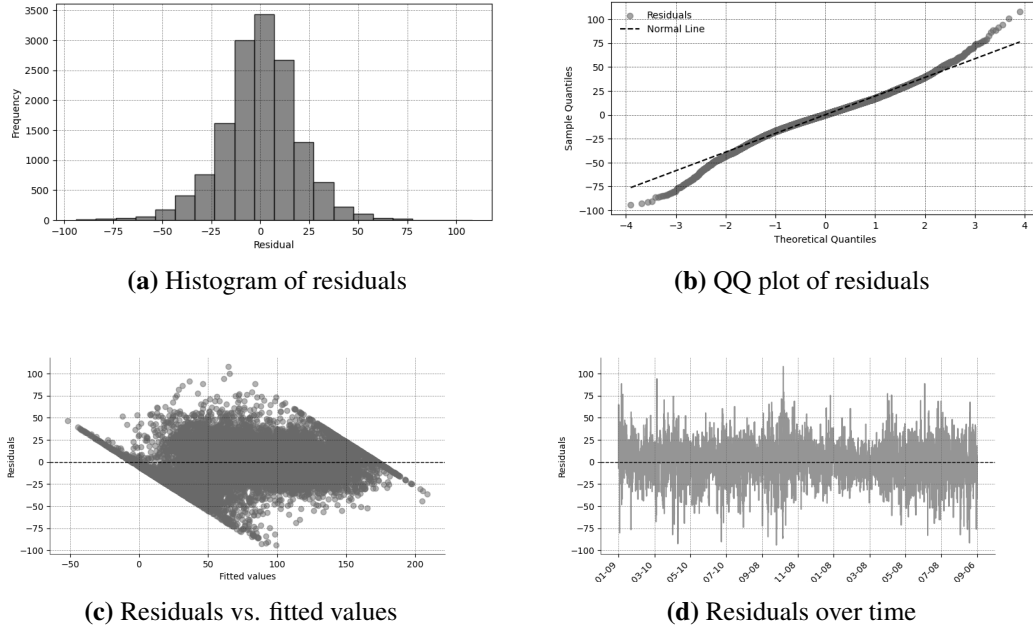


Figure 12: Diagnostic plots for model residuals: (a) QQ plot of residuals, (b) Histogram of residuals, (c) Residuals vs. fitted values, (d) Residuals over time.

In summary, the residual diagnostics indicate that while the model successfully captures broad price movements, key econometric assumptions are not fully satisfied. The presence of non-normal residuals, heteroscedasticity, and autocorrelation suggests that further refinements may be necessary. Potential improvements include incorporating volatility models, such as the GARCH model introduced by Bollerslev (1986), to better capture conditional heteroscedasticity, or introducing additional lagged variables to address autocorrelation. These findings highlight the complexity of electricity price behavior and the need for robust forecasting methodologies capable of handling extreme price variations and time-dependent patterns.

5.1.3 Forecast Evaluation and Limitations

The deterministic forecast generated by the model aligns well with actual day-ahead prices during stable periods, as shown in Figure 13. This forecast is generated for the 24 hourly periods starting from 00:00 to 23:00 of the next day on **September 17, 2024**, leveraging the characteristics and information available prior to the day-ahead market closure. It represents a single deterministic forecast for the entire day-ahead price curve, capturing the hourly dynamics over a full 24-hour period.

However, extreme price fluctuations remain challenging to predict, highlighting areas for improvement, such as introducing non-linear modeling techniques or additional predictors to capture these anomalies effectively. The observed forecasting performance

is not unexpected, as electricity markets exhibit complex price formation mechanisms influenced by demand-supply imbalances, regulatory interventions, and sudden renewable fluctuations.

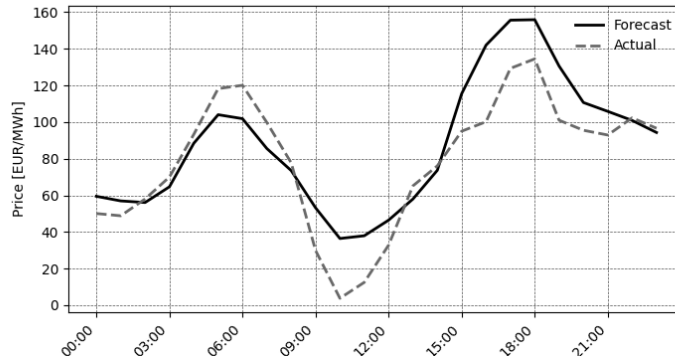


Figure 13: Deterministic forecast vs. actual day-ahead electricity prices on 2024-09-17.

Despite these limitations, the primary objective of this thesis is not solely focused on price forecasting accuracy but rather on exploring stochastic optimization techniques for bidding strategies. The deterministic forecast serves as an input for constructing bidding strategies in electricity markets, where price uncertainties are handled through scenario-based optimization rather than exact point forecasts. Stochastic programming inherently accounts for the probabilistic nature of future price realizations, making the optimization framework robust even if the deterministic forecast does not perfectly capture extreme events.

Thus, while improvements in forecasting accuracy could refine the model, the current forecasting approach remains sufficient for the primary objective of evaluating and optimizing bid strategies under uncertainty in sequential electricity markets.

5.1.4 Scenario Generation for Day-Ahead Market

Electricity prices exhibit strong autocorrelation, volatility clustering, and occasional extreme deviations due to market shocks, supply-demand imbalances, and fluctuations in renewable generation. As highlighted in the residual diagnostics (Section 12), the model's residuals demonstrate heavy tails (Figure 12b), heteroscedasticity (Figure 12c), and significant autocorrelation (Figure 12d). These findings indicate that extreme price movements occur more frequently than a normal distribution would predict, while periods of volatility clustering suggest that large price deviations tend to be followed by further fluctuations. Additionally, the persistence of residuals over time violates the assumption of independent errors, reinforcing the need for a more

sophisticated modeling approach.

To address these limitations and better capture the stochastic variability inherent in electricity prices, this study models the residual component of the forecast using an Ornstein–Uhlenbeck (OU) process, drawing on the methodologies of Benth et al. (2007), Keles et al. (2012), and Meyer-Brandis and Tankov (2008). Unlike models that assume independent noise, the OU process explicitly captures the mean-reverting nature of electricity prices while allowing for stochastic volatility and the inclusion of rare but extreme price jumps. By incorporating this process into the scenario generation framework, the model ensures that deviations from the deterministic forecast align with empirically observed characteristics of electricity markets. This enhances the robustness and realism of the stochastic modeling approach applied to day-ahead price uncertainty.

The scenario generation process begins by decomposing observed market prices into a systematic forecasted component, denoted as $\hat{P}_t^{DA}$, and an unpredictable residual component, ϵ_t , computed as:

$$\epsilon_t = P_t^{DA} - \hat{P}_t^{DA}. \quad (3)$$

Instead of assuming residuals are purely random, the stochastic nature of ϵ_t is explicitly modeled using an OU process, which captures mean reversion — a key characteristic of electricity prices where extreme deviations tend to return to an equilibrium level over time. The continuous-time OU process is defined by the stochastic differential equation:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t + J_t, \quad (4)$$

where θ represents the speed of mean reversion, μ is the long-term equilibrium level of the residuals, σ denotes the volatility of the fluctuations, and dW_t is a standard Wiener process that introduces continuous random variations. To account for rare but extreme price fluctuations, a jump component J_t is included in the Ornstein–Uhlenbeck (OU) process, capturing large deviations from the typical residual behavior. Jumps are empirically derived from the model residuals by identifying observations where the absolute deviation from the mean exceeds three standard deviations of the OU process, i.e., $|\epsilon_t - \mu| > 3\sigma$, following the methodology proposed by Russo et al. (2021). These extreme residuals form the empirical jump distribution. The jump size is modeled using the mean and standard deviation of

these outliers, while jump occurrences are assumed to follow a Poisson process with intensity λ , estimated as the ratio of jump events to the total number of residual observations. This setup ensures that the simulated price scenarios reflect the observed frequency and magnitude of real-world price spikes, thereby preserving the heavy-tailed nature of electricity price distributions in the generated scenarios.

The parameters of the discrete-time approximation of the continuous Ornstein–Uhlenbeck (OU) process are estimated based on historical residuals. The mean-reversion rate θ , volatility σ , and long-term mean μ are calculated as follows:

$$\theta = -\frac{\ln(\rho)}{\Delta t}, \quad (5)$$

$$\sigma = \text{std}(\epsilon) \cdot \sqrt{\frac{2\theta}{1 - \rho^2}}, \quad (6)$$

$$\mu = \text{mean}(\epsilon), \quad (7)$$

where ρ is the lag-1 autocorrelation of the residuals ϵ , and Δt is the time increment (set to 1 hour). These formulas allow for parameter calibration of the OU process in a way that reflects the temporal dependencies and volatility structure observed in historical data.

After parameter estimation, the OU process is simulated over a total period:

$$T = T_{\text{burn-in}} + T_{\text{forecast}}, \quad (8)$$

where $T_{\text{burn-in}}$ allows the process to stabilize before generating 24-hour price paths for the next day. In this study, we assume a burn-in period of 28 days (i.e., 672 hours) for each simulation trial. After this stabilization period, a 24-hour sequence from 00:00 to 23:00 is extracted for each simulated series. These simulated paths represent the stochastic components of day-ahead price uncertainty and are repeated for a total of 1000 Monte Carlo simulations. This approach follows established practices in recent literature on short-term stochastic modeling in electricity markets with high renewable penetration, such as the methodology presented by Zheng, Wang, and Liu (2015).

$$X_{t+1} = X_t + \theta(\mu - X_t)\Delta t + \sigma\xi_t + J_t, \quad (9)$$

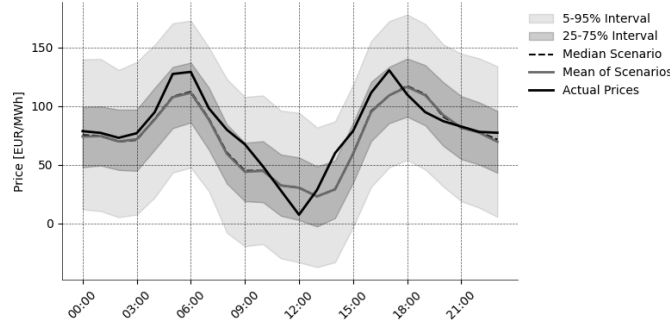
where $\xi_t \sim \mathcal{N}(0, \sqrt{\Delta t})$ represents normally distributed random variations. The final price scenarios are then obtained by adding the simulated residuals back to the

deterministic forecast:

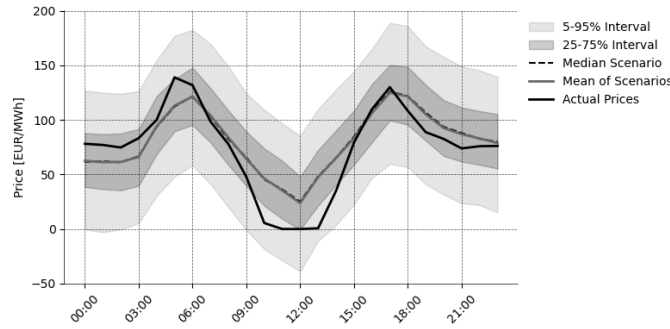
$$P_t^{(n)} = \hat{P}_t^{DA} + X_t^{(n)}, \quad n = 1, \dots, N_s, \quad (10)$$

where $N_s = 1000$ represents the total number of simulated price scenarios.

Figures 14a and 14b illustrate the results for **September 17th** and **18th, 2024**, respectively, depicting the generated day-ahead price scenarios. The figures include 1000 simulated paths, percentile bands (5th–95th and 25th–75th), the median scenario, the mean forecast, and the realized market clearing price (MCP). The percentile bands encapsulate the uncertainty in price evolution, providing a probabilistic range of possible outcomes. The alignment of the mean scenario with the realized MCP indicates that, despite the inherent variability in electricity markets, the hybrid forecasting approach incorporating residual-based scenario generation reasonably captures expected price trends.



(a) Day-ahead electricity price forecast (September 17, 2024).



(b) Day-Ahead Electricity Price Forecast (September 18, 2024).

Figure 14: Comparison of Day-Ahead price forecasts with simulated scenarios for September 17 and 18, 2024.

The study includes price scenario forecasts for the period spanning from **September 17, 2023**, to **October 16, 2023**. However, for brevity, only two representative days are presented here. These two days highlight typical price patterns, including peak and off-peak periods, demonstrating the performance of the model across different market conditions.

For **September 17th** (Figure 14a), the median scenario closely follows the realized MCP, suggesting that the deterministic component of the model effectively captures fundamental market movements. The 25th–75th percentile range remains relatively narrow during stable periods but expands during peak hours, indicating higher uncertainty in price formation during high-demand periods. The 5th–95th percentile band confirms that extreme price events are possible but occur infrequently, as expected.

On **September 18th** (Figure 14b), the general alignment between the mean forecast and the actual MCP is maintained. However, during midday hours, deviations are more pronounced, suggesting that additional refinement in modeling extreme price movements may be necessary. The broader percentile bands during these hours highlight greater uncertainty, likely due to volatile supply-demand conditions or renewable generation fluctuations.

The reduced scenarios, generated via Wasserstein K-Means clustering, effectively summarize the original set of 1,000 simulated day-ahead price paths into ten representative scenarios ($k = 10$). Wasserstein K-Means is a clustering technique that minimizes the Wasserstein distance between probability distributions, ensuring that the reduced scenarios preserve the statistical properties of the original dataset while significantly reducing computational complexity. This method groups similar scenarios into clusters, with each cluster represented by its centroid, and assigns probabilities based on the proportion of original scenarios within each cluster.

The clustering process minimizes the within-cluster variance:

$$\min \sum_{i=1}^k \sum_{j \in C_i} \|x_j - \mu_i\|^2, \quad (11)$$

where C_i is the set of scenarios in cluster i , μ_i is the centroid of cluster i , and x_j are the individual scenarios. The solution ensures that each cluster minimizes the deviation between its members and the cluster centroid.

Probabilities for each cluster are calculated as:

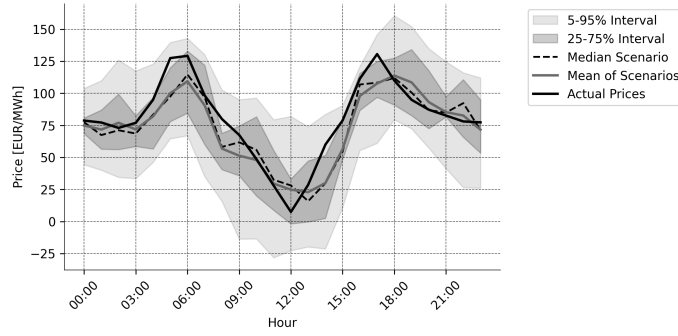
$$p_i = \frac{|C_i|}{N}, \quad (12)$$

where $|C_i|$ is the number of scenarios in cluster i , and N is the total number of scenarios in the original dataset.

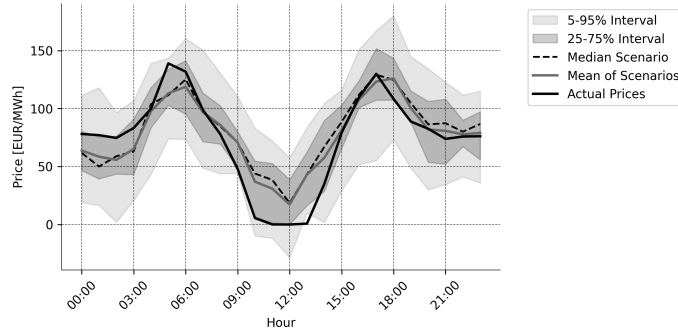
In the context of electricity markets, clustering techniques such as K-Means have been widely applied for scenario reduction to balance accuracy and computational efficiency. For example, Zhao et al. (2021) propose a scenario reduction technique

based on improved K-Means clustering combined with the Simultaneous Backward Reduction algorithm for wind power scenarios. Their method effectively preserves statistical characteristics while significantly reducing dimensionality. Similarly, Ho et al. (2017) introduce a multilevel Wasserstein K-Means approach to simultaneously identify intra-group and inter-group patterns, making it well-suited for high-dimensional stochastic modeling tasks such as those encountered in power systems.

For **September 17th** and **18th**, the reduced scenarios capture the variability and key characteristics of the price distributions, as depicted in Figures 15a and 15b, respectively. Confidence intervals, such as the 5th-95th and 25th-75th percentiles, illustrate the range of price variability, while the mean and median scenarios align closely with observed market prices during stable periods. Table 5 provides the probabilities for each of the ten clusters, demonstrating their representativeness of the original 1000 scenarios.



(a) Reduced Day-Ahead Scenarios (September 17, 2024).



(b) Reduced Day-Ahead Scenarios (September 18, 2024).

Figure 15: Reduced day-ahead price scenarios for September 17 and 18, 2024, using Wasserstein K-Means clustering.

Table 5: Day-Ahead Cluster Probabilities for Reduced Scenarios ($N = 10$).

Cluster	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
(09/17)	0.111	0.155	0.078	0.120	0.073	0.077	0.078	0.072	0.113	0.123
(09/18)	0.103	0.107	0.060	0.119	0.070	0.115	0.074	0.145	0.127	0.080

This clustering approach ensures computational efficiency for downstream optimization tasks, such as scenario tree modeling, without compromising the fidelity of the underlying data distribution. It is particularly useful in electricity markets, where large scenario sets can become computationally prohibitive for real-time decision-making.

In the context of scenario reduction, while the probabilities for each cluster are calculated based on the proportion of original scenarios within each cluster, advanced methods can enhance the representativeness of these probabilities. For instance, Lv et al. (2019) proposed an improved clustering algorithm that incorporates the temporal characteristics of distributed generation outputs, leading to more accurate and informative scenario reductions. Additionally, Ziel (2020) introduced the use of energy distance for scenario reduction, which better preserves the statistical properties of the original dataset compared to traditional distance-based methods. Implementing such advanced techniques could significantly improve the quality of reduced scenarios, thereby enhancing the reliability of downstream optimization results.

5.2 Modeling Uncertainty in the Intraday Market

Uncertainty in the intraday market is modeled using a conditional regression approach that incorporates both day-ahead market variables and intraday-specific predictors. This section describes the data preparation process, the regression model, key results, and residual analysis, culminating in an evaluation of the deterministic forecast.

The regression model incorporates a wide range of predictors to capture the dynamic interplay between day-ahead and intraday markets. Key features include the realized day-ahead price (**DA Price**), residual load forecasts (**RL**), and lagged hourly intraday prices (**ID3**), alongside variables such as a weekend indicator and rolling minimum/maximum prices.

The correlation matrix in figure 16 highlights the relationships between these predictors and the target variable, **ID3 Price**. Notably, strong correlations are observed between lagged intraday prices and the target variable, confirming their predictive significance. Moderate correlations are evident for residual load and day-ahead prices, indicating their influence on intraday price dynamics.

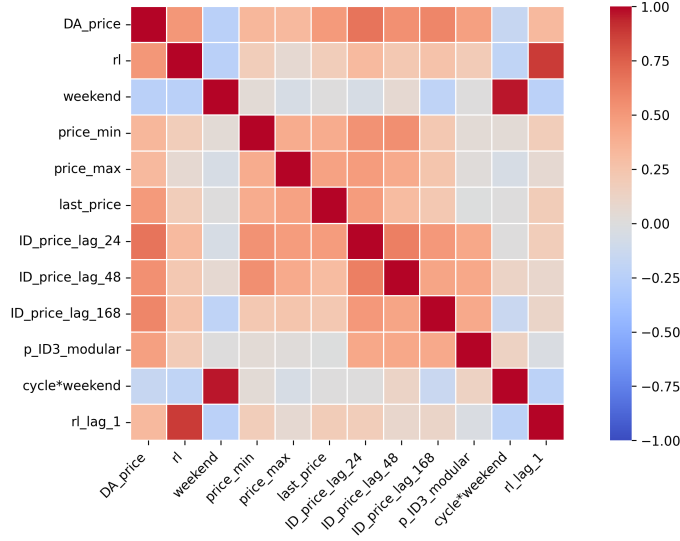


Figure 16: Correlation matrix of intraday market predictors.

The multiple regression equation for intraday prices is defined as follows:

$$\begin{aligned}
 ID3 \text{ Price}_t = & \alpha_0 + \alpha_1 DA \text{ Price}_t + \alpha_2 RL_t + \alpha_3 Weekend \\
 & + \alpha_4 Price \text{ Min}_{t-24} + \alpha_5 Price \text{ Max}_{t-24} + \alpha_6 Last \text{ Price}_{t-1} \\
 & + \alpha_7 ID \text{ Price Lag}_1 + \alpha_8 ID \text{ Price Lag}_2 + \alpha_9 ID \text{ Price Lag}_3 \\
 & + \alpha_{10} Modular \text{ Price} + \alpha_{11} Cycle \text{ Interaction} + \alpha_{12} RL \text{ Lag}_1 + \epsilon_t,
 \end{aligned} \tag{13}$$

where:

- **ID3 Price_t**: Intraday price at time t ,
- **DA Price_t**: Realized day-ahead price,
- **RL_t**: Residual load forecast at time t ,
- **Weekend**: Binary variable indicating weekends,
- **Price Min_{t-24}**: Rolling minimum price over the past 24 hours,
- **Price Max_{t-24}**: Rolling maximum price over the past 24 hours,
- **Last Price_{t-1}**: Intraday price at 23:00 of the previous day,
- **ID Price Lag_n**: Lagged intraday prices at 1, 2, and 3 hour intervals,
- **Modular Price**: Average hourly price shifted by 24 hours,
- **Cycle Interaction**: Interaction term between modular prices and the weekend indicator,
- **RL Lag₁**: 1-period lagged residual load forecast,
- **ε_t**: Error term.

The regression results, presented in Table 9, indicate strong predictive performance for intraday electricity prices, with an R^2 of 0.788. The day-ahead price is the most influential predictor, reflecting the dependence of intraday market dynamics on prior day-ahead commitments. Additionally, residual load and its lagged terms are significant, emphasizing the role of real-time supply-demand imbalances. Lagged intraday prices at 24, 48, and 168 hours further highlight autocorrelation effects, consistent with findings from the day-ahead model (see Section 5.1.1). While price extremes and modular price components contribute to price formation, the weekend indicator is statistically insignificant, suggesting that once other factors are controlled for, weekend effects are minimal.

Despite the model's strong explanatory power, some limitations remain. The condition number suggests potential multicollinearity, though its impact is mitigated by robust standard errors. The Durbin-Watson statistic of 0.545 indicates serial correlation in the residuals, a common issue in electricity price models. As observed in the day-ahead analysis, more advanced time-series techniques, such as stochastic volatility models, could improve forecasts. However, as the primary objective is scenario generation for stochastic optimization rather than perfect forecasting, the current model remains sufficient for the intended application.

Residual diagnostics for the intraday price model, as visualized in Figure 17a through Figure 17d, highlight several departures from classical regression assumptions, including normality, homoscedasticity, and serial independence as highlighted for the day-ahead case in Section 5.1.2.

The histogram in Figure 17a shows that the residuals are approximately symmetric around zero, but not perfectly bell-shaped, indicating mild skewness. This is further reinforced by the QQ plot in Figure 17b, which reveals heavy tails at both ends of the distribution—suggesting that extreme residuals occur more frequently than would be expected under a normal distribution.

Figure 17c (residuals versus fitted values) shows a funnel-like shape, where the spread of residuals varies with the magnitude of fitted values. This pattern is indicative of heteroscedasticity, meaning the error variance is not constant and tends to increase with higher predicted prices. Such heteroscedasticity violates OLS assumptions and can bias inference unless corrected, for instance by using robust standard errors or volatility models like GARCH.

Temporal dynamics of residuals are shown in Figure 17d. Clear bands of high and low volatility are visible, indicating volatility clustering—a hallmark of time series exhibiting conditional heteroscedasticity. Moreover, residuals appear to follow a time-dependent structure rather than white noise. This is quantitatively confirmed by the Ljung–Box test, which yielded a test statistic of 18,376 with a p-value of 0.0014, strongly rejecting the null hypothesis of no autocorrelation. These findings suggest that residuals are serially correlated and that past forecast errors influence future ones—a violation of the classical assumption of error independence.

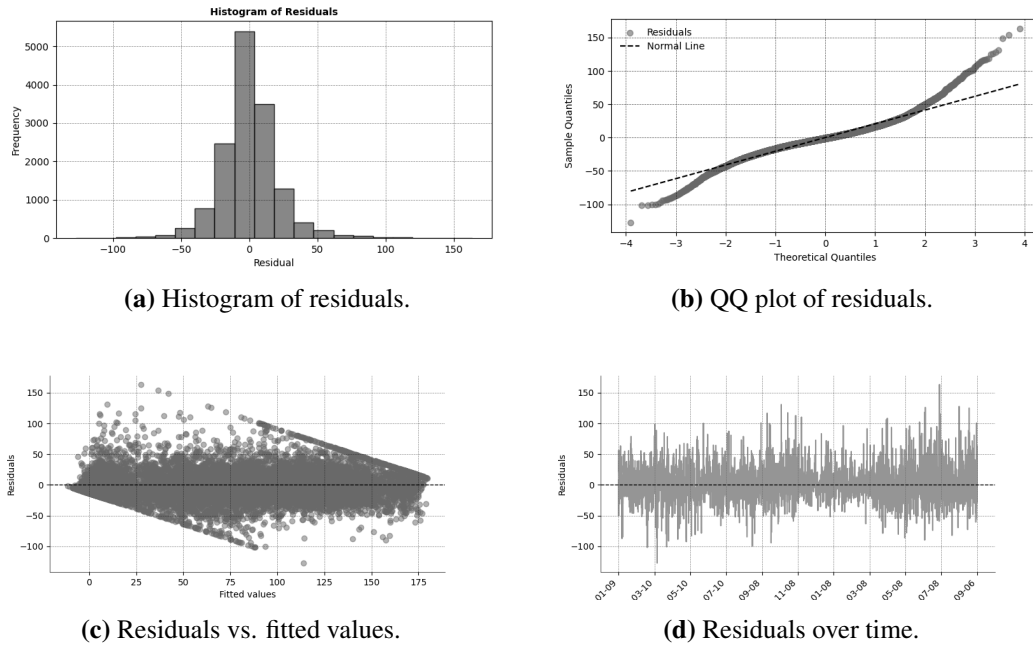


Figure 17: Diagnostic plots for intraday model residuals .

Overall, while the model performs reasonably well in capturing broad market behavior, these residual patterns reinforce the importance of augmenting the model with dynamic volatility and error correction mechanisms to ensure robustness in real-world trading applications.

The deterministic forecast for intraday prices, shown in Figure 18, closely follows actual market prices under stable conditions but demonstrates notable discrepancies during volatile periods. Given that the day-ahead market-clearing price (MCP) is unknown at the time of intraday price forecasting, the model replaces it with the forecasted day-ahead price scenario to maintain consistency. This adjustment ensures that each intraday scenario is built upon a coherent day-ahead forecast, reflecting realistic market conditions.

Overall, the intraday price model exhibits strong predictive performance in stable



Figure 18: Deterministic Intraday ID3 Price Forecast vs. Actual Prices on 2024-09-17.

market conditions, with residual patterns largely aligning with expectations.

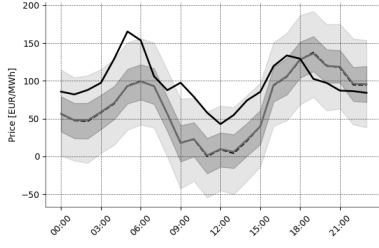
5.2.1 Conditional Scenario Generation in the Intraday Market

Intraday electricity price scenarios are generated conditional on the day-ahead market outcomes. Since the day-ahead bidding process determines the initial market commitments, intraday prices reflect adjustments made in response to forecast errors, unforeseen fluctuations in supply and demand, and short-term market dynamics. The intraday market thus exhibits greater volatility and responsiveness compared to the day-ahead market, necessitating a scenario-based approach to capture potential deviations from prior commitments.

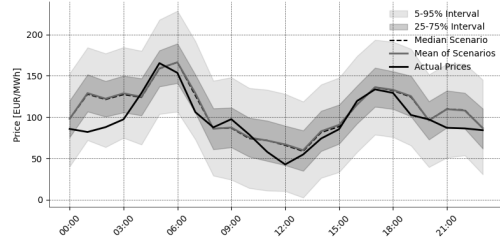
As detailed in Section 5.1.4, the day-ahead price scenarios form the basis for subsequent intraday price simulations. Given the high dimensionality of day-ahead scenarios, we use a reduced set of day-ahead scenarios obtained through scenario clustering ($k = 10$). Each of these reduced day-ahead scenarios serves as the foundation for generating 1000 intraday price simulations. This two-stage approach ensures computational efficiency while maintaining the variability inherent in real-world price fluctuations.

For each reduced day-ahead scenario, we construct a set of 10 distinct intraday nodes, where each node consists of 1000 simulated price paths. This results in a structured scenario tree that captures potential intraday deviations while preserving the statistical distribution of the original dataset. Due to the large number of scenarios generated, visualizing all 10 intraday nodes per day is impractical. Instead, Figures 19 present two representative intraday nodes per day, providing insight into the variability across different scenarios while avoiding redundancy.

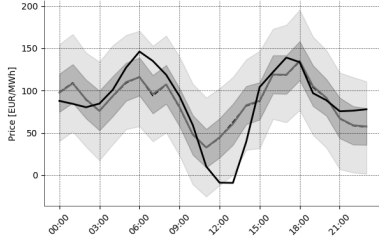
The intraday price forecast scenarios depicted in Figure 19 illustrate price uncertainty



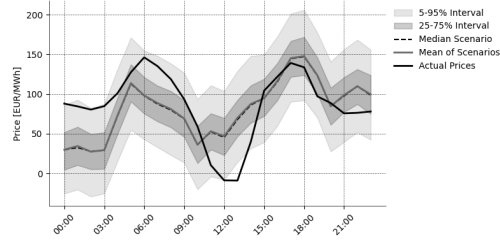
(a) Intraday price forecast scenarios conditional on the first day-ahead scenario (09-17-2024).



(b) Intraday price forecast scenarios conditional on the second day-ahead scenario (09-17-2024).



(c) Intraday price forecast scenarios conditional on the first day-ahead scenario (09-18-2024).



(d) Intraday price forecast scenarios conditional on the second day-ahead scenario (09-18-2024).

Figure 19: Intraday price forecast scenarios for September 17 and 18, 2024, conditional on their respective day-ahead scenarios.

conditioned on different day-ahead market realizations. The figures show simulated price evolution throughout the intraday market trading hours. The percentile bands indicate the range of price fluctuations, while the mean and median trajectories highlight the expected price path. Notably, deviations are more pronounced during early morning and peak demand hours, reflecting the impact of real-time market adjustments and sudden supply changes.

To ensure computational feasibility while preserving key price characteristics, scenario reduction techniques are applied to the intraday simulations. Following the clustering approach described in Section 5.1.4, the 1000 simulated intraday scenarios per day-ahead scenario are reduced to $k = 10$ clusters using k -means clustering.

Table 6: Cluster Probabilities for Reduced Intraday Scenarios ($N = 10$).

Cluster	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
(09/17 Scenario 0)	0.097	0.073	0.091	0.125	0.104	0.101	0.067	0.131	0.119	0.092
(09/17 Scenario 1)	0.067	0.112	0.113	0.091	0.088	0.099	0.121	0.123	0.109	0.077
(09/18 Scenario 0)	0.094	0.092	0.088	0.098	0.112	0.127	0.081	0.095	0.099	0.114
(09/18 Scenario 1)	0.085	0.129	0.098	0.120	0.113	0.104	0.075	0.069	0.096	0.111

The reduced intraday price forecasts shown in Figure 20 represent condensed versions of the original 1,000 simulations while preserving key market characteristics. The clustering approach ensures that critical patterns—such as price peaks and troughs—remain

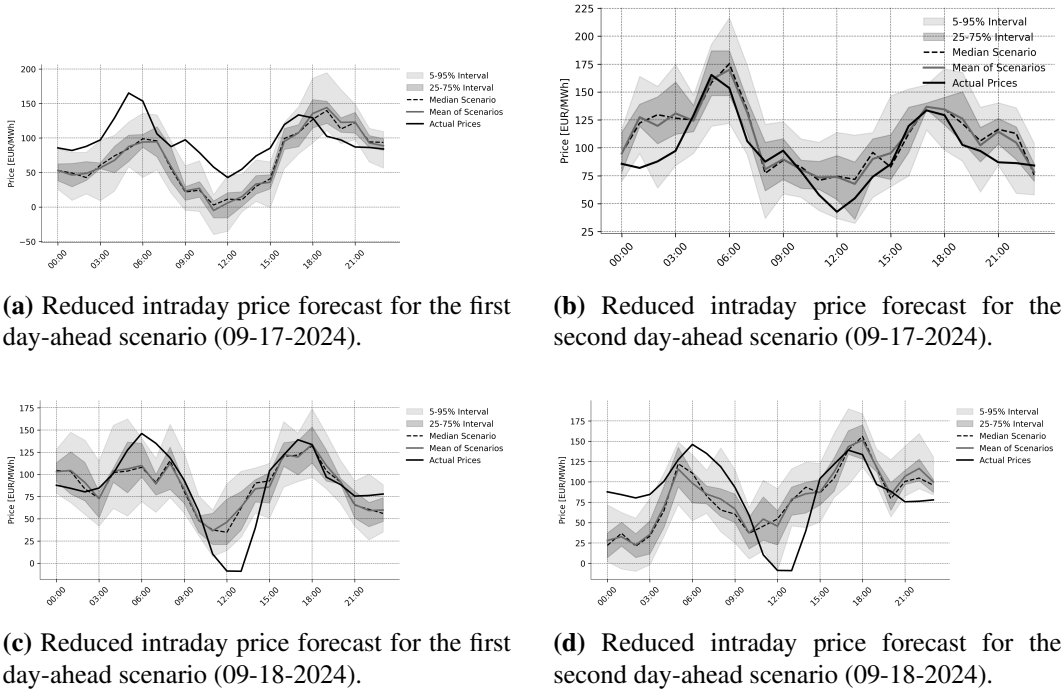


Figure 20: Comparison of reduced intraday price forecast scenarios for September 17 and 18, 2024, conditioned on their respective day-ahead scenarios.

evident, particularly during high-demand hours. Notably, early morning and late evening price swings are retained in the reduced scenarios, maintaining consistency with typical intraday market behavior. These reductions enhance computational efficiency without sacrificing the statistical representativeness required for reliable decision-making. Additionally, Table 6 displays the cluster probability distributions for two representative intraday scenarios out of the ten total clusters for the two days as above, demonstrating the alignment of reduced scenarios with the original distribution.

By structuring the reduced intraday scenarios in this manner, the model effectively balances computational feasibility with accuracy in forecasting. The scenario reduction method enables effective decision-making in intraday market participation, allowing traders and operators to anticipate price trends with a well-structured probabilistic representation.

5.3 Scenario Tree Representation

The scenario tree representation forms a foundational element of the stochastic Mixed-Integer Linear Programming optimization model used for bid ladder formation in electricity wholesale spot markets. It enables the structured modeling of uncertainty in both the day-ahead and intraday markets by discretizing potential price realizations

into a probabilistic framework. The scenario tree adopts a hierarchical structure: each reduced day-ahead scenario branches into ten conditional intraday scenarios, forming a complete 10×10 scenario tree. This approach strikes a balance between comprehensive uncertainty representation and computational tractability.

For the methodology to deliver meaningful results, it is essential to accurately characterize the uncertainty present in real-world electricity markets. In the Dutch market context, this uncertainty is represented as a finite probability space Ω , defined by discrete scenarios ω structured in a three-stage scenario tree: initial deterministic inputs i , day-ahead price uncertainty j , and intraday price uncertainty k . The first stage reflects price uncertainty in the day-ahead market, primarily driven by updates in renewable generation and demand forecasts that occur between the reserve market gate closure (around 10:00 CET, D-1) and the day-ahead market gate closure (12:00 CET, D-1). The second stage captures intraday uncertainty, based on updated residual load and renewable generation forecasts available approximately 60 minutes before delivery. This leads to price uncertainty in the intraday market, particularly in the ID3 index, which reflects market conditions during the final hours before delivery.

Figure 21 illustrates the overall structure of the scenario tree. Each node in the day-ahead market represents a clustered forecast of day-ahead prices, which further branches into ten distinct intraday scenarios. This hierarchical branching captures the sequential nature of price evolution and allows for adaptive trading strategies in response to market updates. As such, the tree serves as a discrete probabilistic representation of possible future market developments, effectively modeling the uncertainty inherent in liberalized electricity wholesale spot markets.

Figures 35 and 36 present the detailed structure of the scenario trees for **September 17 and 18, 2024**, respectively. Each subfigure includes a full realization of the scenario tree for the respective trading day, incorporating both the clustered day-ahead forecasts and their corresponding intraday price trajectories. The solid black line in each subfigure represents the realized market price, while the orange dashed lines indicate the simulated scenario paths. These trajectories demonstrate substantial variation, particularly at the upper and lower bounds, capturing both typical fluctuations and extreme price movements.

The clustering process ensures that each node represents a statistically robust price

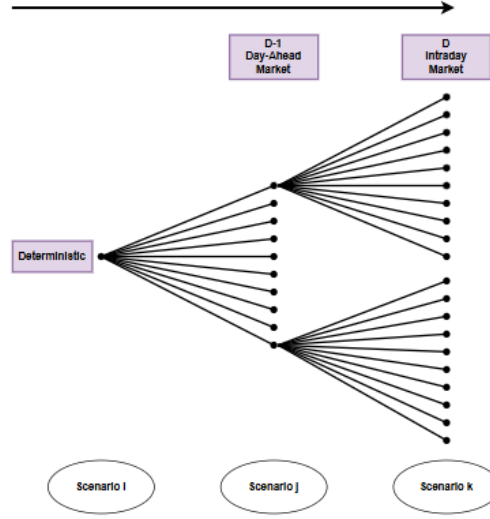


Figure 21: Illustration of the 10x10 scenario tree structure. Each day-ahead node branches into ten conditional intraday nodes, representing uncertainty in both markets.

path, preserving key market features such as volatility clustering, mean reversion, and price spikes. Each scenario tree comprises 101 nodes and is replicated for each of the 30 days between **September 17** and **October 16, 2024**.

Tables 5 and 6 provide the probability distributions associated with the day-ahead and intraday scenarios, respectively. These distributions reflect the likelihood of each scenario occurring. For example, on **September 17**, the day-ahead scenario probabilities range from 0.073 to 0.155, and the corresponding conditional intraday probabilities span from 0.067 to 0.131 for the two representative clusters out of the total ten. A similar range is observed for **September 18**, highlighting the inherent uncertainty of electricity price formation.

The median scenario in each tree aligns well with the realized trend, reflecting the accuracy of the clustering methodology. Percentile bands (5th–95th and 25th–75th) encapsulate expected variability, enabling the optimization framework to account for non-Gaussian characteristics such as heavy tails and jumps.

From a market dynamics perspective, the scenario trees highlight the interaction between day-ahead commitments and intraday adjustments. The probability-weighted structure supports informed trading decisions, where higher-probability paths represent expected trends, and lower-probability paths account for tail risks and market shocks.

The scenario tree is mathematically represented as:

$$T = \bigcup_{i=1}^{N_{DA}} \bigcup_{j=1}^{N_{ID}} \{S_i^{DA}, S_{i,j}^{ID}, p_{i,j}\}^5 \quad (14)$$

where:

- S_i^{DA} : Day-ahead scenario i ,
- $S_{i,j}^{ID}$: Intraday scenario j conditional on day-ahead scenario i ,
- $p_{i,j}$: Probability of the joint scenario $(S_i^{DA}, S_{i,j}^{ID})$,
- N_{DA} : Total number of day-ahead scenarios,
- N_{ID} : Total number of intraday scenarios per day-ahead scenario.

The joint scenario probabilities are computed as:

$$p_{i,j} = p_i^{DA} \cdot p_{j|i}^{ID}, \quad (15)$$

where p_i^{DA} is the marginal probability of the i -th day-ahead scenario, and $p_{j|i}^{ID}$ is the conditional probability of the j -th intraday scenario given that day-ahead realization.

This formulation ensures alignment with real-world market behavior by capturing both central trends and edge-case outcomes. The use of clustering-based scenario reduction combined with probability-weighted branching enables efficient and realistic modeling. When integrated into the stochastic Mixed-Integer Linear Programming framework, this scenario tree allows for robust, risk-aware decision-making that helps market participants hedge against volatility while maximizing revenue opportunities.

In summary, the scenario tree structure illustrated in Figures 35 and 36 offers a practical and powerful framework for modeling electricity price uncertainty. Its hierarchical, probabilistic nature supports both flexibility and precision in developing resilient bid strategies under uncertainty.

6 Two-Stage Stochastic Optimization Model

This chapter describes the two-stage stochastic optimization model, designed to optimize bidding strategies for electricity wholesale spot markets. The model considers both the expected contribution margin and Conditional Value at Risk (CVaR), balancing profitability and risk aversion.

The optimization incorporates uncertainties from day-ahead and intraday markets

⁵ This expression defines the full scenario tree T as the union of all possible combinations of day-ahead scenarios S_i^{DA} and their corresponding conditional intraday scenarios $S_{i,j}^{ID}$, each paired with a joint probability $p_{i,j}$. The outer and inner unions reflect the hierarchical structure, where each day-ahead scenario branches into multiple intraday realizations.

using a 10x10 scenario tree representation. The objective is to maximize profits while managing risks associated with price and production variability. The details of the model components, constraints, and results are outlined below.

6.1 Objective Function

The objective function maximizes the risk-adjusted expected contribution margin while incorporating Conditional Value at Risk (CVaR) and penalties for deviations in bidding decisions:

$$\max \left[(1 - \lambda) \mathbb{E}[\pi] + \lambda \cdot \text{CVaR} - w \sum_{h=1}^{24} \sum_{i=1}^I (x_{i,h}^{\text{DA}} - \psi^{\text{DA}})^2 - w \sum_{h=1}^{24} \sum_{i=1}^I \sum_{j=1}^J (x_{i,j,h}^{\text{ID}} - \psi^{\text{ID}})^2 \right] \quad (16)$$

Where:

- λ : Risk aversion parameter ($0 \leq \lambda \leq 1$).
- $\mathbb{E}[\pi]$: Expected contribution margin.
- CVaR: Conditional Value at Risk (downside risk measure).
- w : Penalty weight for deviation from the limit price.
- $\psi^{\text{DA}}, \psi^{\text{ID}}$: Limit prices for DA and ID markets.
- $x_{i,h}^{\text{DA}}, x_{i,j,h}^{\text{ID}}$: Committed volumes in DA and ID markets.

6.1.1 Expected Contribution Margin

$$\mathbb{E}[\pi] = \mathbb{E}[\rho] - \mathbb{E}[\kappa]^6 \quad (17)$$

Where:

- $\mathbb{E}[\rho]$: Expected total revenue from DA and ID markets.
- $\mathbb{E}[\kappa]$: Expected total cost based on limit prices.

⁶ Notation Summary:

- I : Number of day-ahead scenarios with probabilities π_i^{DA} .
- J : Number of intraday scenarios per day-ahead scenario with conditional probabilities $\pi_{j|i}^{\text{ID}}$.
- $P_{i,h}^{\text{DA}}$: Day-ahead market price in scenario i at hour h .
- $P_{i,j,h}^{\text{ID}}$: Intraday market price in scenario j (conditional on i) at hour h .
- $x_{i,h}^{\text{DA}}$: Day-ahead commitment in scenario i at hour h .
- $x_{i,j,h}^{\text{ID}}$: Intraday commitment in scenario j under day-ahead scenario i at hour h .
- $\psi^{\text{DA}}, \psi^{\text{ID}}$: Limit prices (e.g., marginal costs or price floors).

6.1.2 Expected Total Revenue

$$\mathbb{E}[\rho] = \underbrace{\sum_{i=1}^I \pi_i^{\text{DA}} \sum_{h=1}^{24} P_{i,h}^{\text{DA}} \cdot x_{i,h}^{\text{DA}}}_{\text{Expected Day-Ahead Revenue}} + \underbrace{\sum_{i=1}^I \pi_i^{\text{DA}} \sum_{j=1}^J \pi_{j|i}^{\text{ID}} \sum_{h=1}^{24} P_{i,j,h}^{\text{ID}} \cdot x_{i,j,h}^{\text{ID}}}_{\text{Expected Intraday Revenue}} \quad (18)$$

6.1.3 Expected Total Cost

$$\mathbb{E}[\kappa] = \underbrace{\sum_{i=1}^I \pi_i^{\text{DA}} \sum_{h=1}^{24} \psi_{i,h}^{\text{DA}} \cdot x_{i,h}^{\text{DA}}}_{\text{Expected Day-Ahead Cost}} + \underbrace{\sum_{i=1}^I \pi_i^{\text{DA}} \sum_{j=1}^J \pi_{j|i}^{\text{ID}} \sum_{h=1}^{24} \psi_{i,j,h}^{\text{ID}} \cdot x_{i,j,h}^{\text{ID}}}_{\text{Expected Intraday Cost}} \quad (19)$$

6.1.4 Conditional Value at Risk (CVaR)

$$\text{CVaR} = \eta - \frac{1}{1 - \alpha} \sum_{i=1}^I p_{\text{DA},i} \cdot s_i \quad (20)$$

Subject to:

$$s_i \geq \eta - \pi_i, \quad s_i \geq 0, \quad \forall i \quad (21)$$

Where:

- η : Value at Risk (VaR) at confidence level α .
- s_i : Shortfall (loss below η) in day-ahead scenario i .
- π_i : Contribution margin under scenario i .

6.2 Constraints

6.2.1 Forecast Constraint

$$\sum_{i=1}^I x_{i,h}^{\text{DA}} + \sum_{i=1}^I \sum_{j=1}^J x_{i,j,h}^{\text{ID}} \leq G_h. \quad (22)$$

This ensures that the total traded volume in the day-ahead and intraday markets does not exceed the available forecasted generation for hour h . Since market participants cannot sell more electricity than expected production, this constraint preserves physical feasibility and prevents overcommitting.

6.2.2 Minimum and Maximum Commitment Constraints

$$\theta_{\min} G_h \leq \sum_{i=1}^I x_{i,h}^{\text{DA}} + \sum_{i=1}^I \sum_{j=1}^J x_{i,j,h}^{\text{ID}} \leq \theta_{\max} G_h. \quad (23)$$

This constraint ensures that the total bidding volume (DA + ID) remains within an acceptable range of forecasted generation G_h . $\theta_{\min}$ represents the minimum

fraction of forecasted generation that must be committed to bids (e.g., to avoid under-utilization). $\theta_{\max}$ sets an upper bound, preventing bids that exceed production capacity significantly.

6.2.3 Minimum Day-Ahead Bids

$$\gamma_{\text{DA}}^{\min} \cdot G_h \leq \sum_{i=1}^I x_{i,h}^{\text{DA}} \leq \gamma_{\text{DA}}^{\max} \cdot G_h, \quad (24)$$

This constraint ensures that the total commitment to the day-ahead market lies within a predefined range, typically between $\gamma_{\text{DA}}^{\min}$ and $\gamma_{\text{DA}}^{\max}$ of the forecasted generation G_h . It reflects the common practice of hedging by securing a baseline share of expected generation in the day-ahead market, while retaining flexibility to adjust remaining volumes in the intraday market based on updated forecasts and market conditions.

6.2.4 Bidding Limit Constraints (with Binary Decision Variables)

We introduce binary variables $z_{i,h}^{\text{DA}} \in \{0, 1\}$ and $z_{i,j,h}^{\text{ID}} \in \{0, 1\}$ that indicate whether a bid is accepted in each scenario and hour. The following big- M constraints ensure that volumes are only committed when prices exceed the respective limit prices:

$$x_{i,h}^{\text{DA}} \leq M \cdot z_{i,h}^{\text{DA}}, \quad z_{i,h}^{\text{DA}} \in \{0, 1\} \quad (25)$$

$$P_{i,h}^{\text{DA}} - \psi^{\text{DA}} \geq -M(1 - z_{i,h}^{\text{DA}}) \quad (26)$$

$$x_{i,j,h}^{\text{ID}} \leq M \cdot z_{i,j,h}^{\text{ID}}, \quad z_{i,j,h}^{\text{ID}} \in \{0, 1\} \quad (27)$$

$$P_{i,j,h}^{\text{ID}} - \psi^{\text{ID}} \geq -M(1 - z_{i,j,h}^{\text{ID}}) \quad (28)$$

Here, M is a sufficiently large constant to deactivate volume decisions when prices are not favorable. These constraints enforce that electricity is only sold in scenarios where the price exceeds the marginal cost or the predefined bid threshold.

6.2.5 CVaR Constraint

$$s_i \geq \eta - \pi_i, \quad s_i \geq 0, \quad \forall i. \quad (29)$$

This constraint ensures that the shortfall variable s_i properly captures downside risk by forcing it to account for losses beyond the VaR threshold η . If a scenario's profit π_i is below the risk threshold η , the shortfall s_i quantifies this deviation. If a scenario's profit is higher than η , then $s_i = 0$, meaning no additional risk exposure.

6.3 Final Model Summary

The constraints collectively ensure that bidding decisions are aligned with the available generation capacity G_h , preventing overcommitment, while risk exposure is effectively managed through CVaR constraints to mitigate potential financial losses. Additionally, market behavior remains realistic by restricting loss-making trades and ensuring economically viable bidding strategies. As a result, the optimization model achieves a balanced trading strategy that maximizes profitability while effectively managing risks in both the day-ahead and intraday electricity markets.

7 Evaluation of Trading Strategies

In this study, trading strategies are evaluated by optimizing bid ladders under distinct risk preferences, characterized by a set of risk aversion parameters (0%, 10%, 25%, 50%) and Conditional Value at Risk (CVaR)⁷ confidence levels (85%, 90%, 95%). This results in nine unique strategy combinations. The risk-neutral strategy is defined by the parameter set (0%, 85%), while the most risk-averse strategy corresponds to (50%, 95%). The optimization framework systematically explores these risk configurations to assess their impact on bidding behavior and trading performance across electricity markets. To illustrate the methodology, we present a case study using two trading test days, **September 17** and **September 18, 2024** instead of the whole extensive 30 day testing sample between **September 17** and **October 16, 2024**.

The selection of these parameter values is guided by the need to balance market realism, risk management, and flexibility in bid allocation between the day-ahead and intraday markets. A cluster size of $k = 10$ is chosen to ensure sufficient granularity in the stochastic optimization model, effectively capturing market uncertainties while maintaining computational feasibility. Additionally, a limit price of -2 €/MWh is introduced to prevent bids at extreme low-price levels, mitigating reliance on outlier market conditions that may not align with practical trading behavior. Specifically, in the context of the HKZ segment in the Netherlands, this limit price represents the threshold for profitability, incorporating the effects of subsidies. This negative price influences bidding behavior by incentivizing production in certain conditions where

⁷ CVaR is a coherent risk metrics, which satisfies the conditions of monotonicity, sub-additivity, homogeneity, and translational invariance.

payments are received for generating electricity.

To establish a baseline revenue stream and mitigate exposure to price fluctuations, 50% of the forecasted generation is committed at the limit price, providing a structured risk-hedging mechanism before engaging in price-dependent bidding — a strategy made viable by the fact that the market clearing price (MCP) is most often above -2 €/MWh. Furthermore, a penalty weight of 0.1 is incorporated to discourage significant deviations from optimal bid allocations, thereby ensuring stability and practical implementability in real-world trading scenarios.

Operational constraints are enforced through forecast thresholds, which are set between 90% and 110% of the forecasted generation ⁸. These constraints ensure that bid commitments remain within reasonable limits, preventing excessive under- or over-commitment while maintaining flexibility. Additionally, day-ahead market bid commitments are restricted to a range of 50% to 90% of the forecasted generation, allowing for strategic flexibility by preserving capacity for participation in the intraday market ⁹. Risk-neutral strategies tend to allocate a higher share of commitments in the day-ahead market, whereas risk-averse strategies maintain greater flexibility for intraday trading, enabling adjustments based on price and volume uncertainties. This structure acknowledges that while the day-ahead market generally exhibits lower volatility, excessive early commitments can lead to inefficiencies if deviations from forecasts occur. By carefully defining these parameters, the optimization framework ensures a structured yet adaptable bidding approach that balances revenue maximization with risk mitigation while adhering to market constraints. These parameters are fully adjustable and have been selected based on a validation process aimed at minimizing volume risk and enhancing trading performance.

Before examining the decision variables and the rationale behind different trading strategies, we first assess the objective function values and overall performance. To quantify the effect of risk considerations, we compute the expected contribution margin and CVaR as averages across all scenario trees each representing a different day. The efficient frontier is then constructed as the set of non-dominated strategies,

⁸ As detailed in Section 6.4.2, operational constraints are enforced by requiring that the total bids in the day-ahead and intraday markets for each hour lie within 90% to 110% of the forecasted generation, i.e., $\theta_G^{\min} = 0.9$ and $\theta_G^{\max} = 1.1$.

⁹ As detailed in Section 6.4.3, day-ahead market bid commitments for each scenario and hour are constrained to lie between 50% and 90% of the forecasted generation. This range, defined by $\gamma_{DA,G,h}^{\min} = 0.5$ and $\gamma_{DA,G,h}^{\max} = 0.9$, enables strategic flexibility by preserving capacity for participation in the intraday market.

ensuring optimal trade-offs between expected returns and risk exposure, such that no alternative strategy achieves a higher return at a lower level of risk.

Figure 22 illustrates the efficient frontier for all combinations of α and λ , along with the risk-neutral strategy. Across all investigated CVaR confidence levels (α), a consistent concave pattern emerges. Table 10 provides the numbers determining the efficient frontier. As expected, incorporating risk aversion (i.e., increasing λ) leads to a reduction in expected contribution margins while simultaneously improving downside risk, as reflected by higher CVaR values. A notable observation from Figure 22 is the significant impact of increasing the risk aversion parameter λ from risk-neutral (0%) to 10%, as well as from 10% to 25%. The first transition results in a substantial decline in the expected contribution margin, decreasing by approximately 8,000–8,545 EUR/day. Simultaneously, the Conditional Value at Risk (CVaR) increases notably by 15,000–18,200 EUR/day, indicating improved resilience under adverse scenarios.

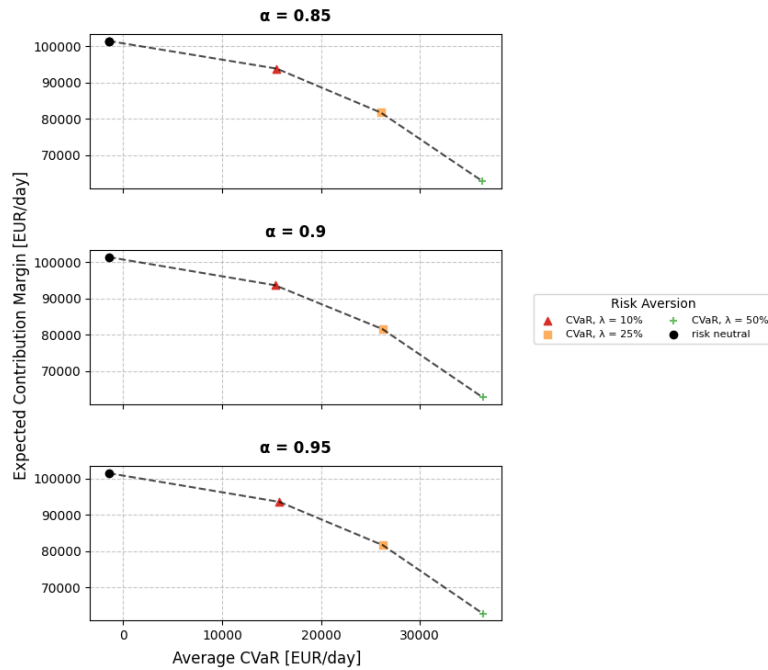


Figure 22: Efficient frontier of trading decisions for different levels of CVaR interval (α) and risk aversion (λ)

The transition from 10% to 25% risk aversion leads to an even more pronounced drop in expected contribution margin, ranging from 11,000 to 12,250 EUR/day. However, this adjustment further mitigates downside risk, as indicated by a corresponding increase in CVaR. Interestingly, the shift from 25% to 50% risk aversion produces the most pronounced effect, as reflected in the efficient frontier of trading decisions.

This substantial increase in risk aversion results in a sharper decline in expected contribution margins, emphasizing the trade-off between profitability and risk mitigation. Across all confidence levels ($\alpha = 0.85, 0.9, 0.95$), the efficient frontier demonstrates a clear inverse relationship between CVaR and expected contribution margin, reinforcing the impact of risk preferences on trading strategies.

Overall, the efficient frontier maintains a concave shape across all α levels. This confirms that none of the defined risk strategies strictly dominates another. In practice, a trader must select a risk preference based on their tolerance and understanding of the trade-off between expected returns and risk exposure.

The figures 23 and 24 illustrate the optimal bidding strategies for a risk-neutral ($\alpha = 0$, CVaR = 0.85) and a risk-averse ($\alpha = 0.5$, CVaR = 0.95) trader on the day-ahead market for two consecutive days, **September 17** and **September 18, 2024**. The scenario leaf probabilities of the bids are visualized using color gradation, while the bidding volumes are represented by the size of the markers. It should be noted that bids submitted on the day-ahead market do not anticipate realizations in the intraday market, and are thus consistent over the represented scenario tree. The underlying distinction between the two strategies is evident in the way bids are distributed across different price levels and scenario probabilities. A key difference compared to the bidding strategies defined in Russo et al. (2021) where long positions are permitted in the day-ahead market is that, in such cases, risk-neutral strategies tend to take long positions in the day ahead market, while risk-averse strategies lean towards sell bids to hedge uncertainties and secure revenues early on. However, in the present analysis, long positions are prohibited in the day-ahead market due to the underlying assumption of generation-based market participation. Consequently, the risk-neutral strategy maximizes its expected contribution margin by allocating more volume to high probability price scenarios, aiming to capture higher revenues at these favorable price points.

For the risk-neutral strategy ($\alpha = 0$, CVaR = 0.85), the trader tends to place bids selectively in higher-probability price scenarios from the scenario tree. This behavior suggests an optimization process aimed at maximizing expected contribution margins by bidding only in price scenarios where the probability of acceptance is high. Consequently, fewer bids are placed in low-probability price scenarios, indicating that the trader does not actively hedge against more uncertain outcomes.

This approach aligns with the risk-neutral objective of maximizing expected returns without significant concern for downside risk.

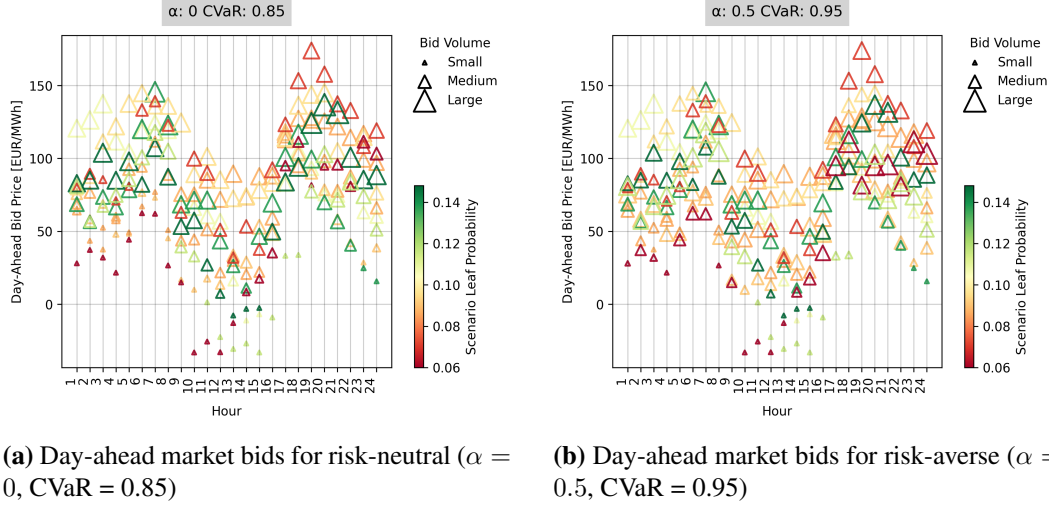


Figure 23: Comparison of day-ahead market bids under risk-neutral and risk-averse strategies on 2024-09-17

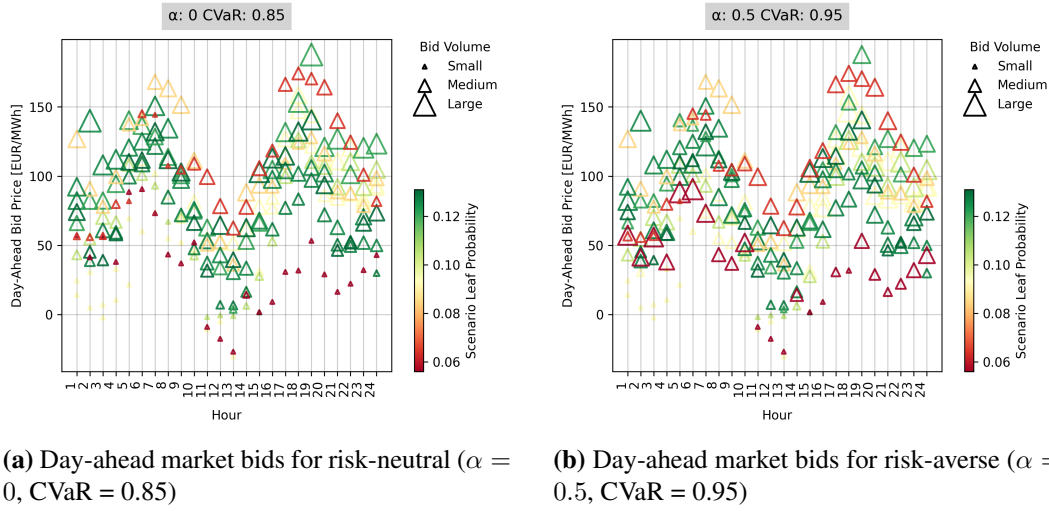


Figure 24: Comparison of day-ahead market bids under risk-neutral and risk-averse strategies on 2024-09-18

In contrast, the risk-averse strategy ($\alpha = 0.5$, CVaR = 0.95) demonstrates a more diversified bidding pattern across multiple price scenarios. The risk-averse agent appears to distribute its bids more broadly, ensuring that trades occur under a variety of market conditions. This strategy is particularly effective in mitigating the volume risk associated with over-reliance on a single market scenario. By spreading bids across a wider range of potential prices, the trader reduces the likelihood of missing opportunities in case the realized market price deviates from the most probable scenarios.

This divergence in strategies has significant implications for market positioning.

The risk-neutral trader's preference for bidding in high-probability scenarios suggests a willingness to engage with intraday markets if the expected day-ahead clearing prices are suboptimal. This aligns with the broader expectation that intraday markets, while more volatile, often provide higher price opportunities. On the other hand, the risk-averse trader seeks to secure a larger share of transactions in the day-ahead market, prioritizing stable and predictable revenues over potential speculative gains in the intraday market.

An interesting pattern observed in both figures is the response to peak electricity demand hours. The risk-averse strategy places relatively higher volumes of bids during peak demand periods, likely as a precautionary measure to lock in favorable prices ahead of time. This contrasts with the risk-neutral strategy, which appears to take a more opportunistic approach, bidding aggressively only in scenarios with a high expected return.

Overall, these findings illustrate how varying risk attitudes result in fundamentally different bidding behaviors in sequential electricity markets. The risk-neutral strategy prioritizes opportunistic, high-margin trades, often at the expense of bid acceptance — particularly when bid prices exceed the eventual market-clearing price. In contrast, the risk-averse approach prioritizes bid security, favoring stable revenue streams even if it means accepting lower expected profits. These distinctions offer valuable insights into how traders navigate the trade-off between profitability, risk exposure, and evolving market conditions in electricity trading.

Figures 25 and 26 present the resulting contribution margins from the day-ahead and intraday markets for the considered strategies across the entire trading test sample from **September 17, 2024**, to **October 16, 2024**, across different levels of risk aversion. A key observation is that the highest contribution margin in the day-ahead market is achieved by the risk-neutral strategy. This result is expected, as the risk-neutral trader prioritizes maximizing expected returns by allocating higher quantities to high-probability price scenarios in the day-ahead market. By doing so, they secure revenues through optimal bid placement, leading to high but volatile income. The color scheme in the figures reflects the transition from risk-neutral to increasingly risk-averse strategies, with deepest red representing the most risk-neutral case and progressively greener tones indicating higher levels of risk aversion.

Conversely, the most risk-averse strategy exhibits the lowest contribution margin

across the trading days. This outcome arises from a more conservative bidding approach, characterized by spreading bids across a wider range of price scenarios to mitigate downside risk. While this reduces exposure to adverse price fluctuations, it also limits potential gains, resulting in lower overall contribution margins compared to less risk-averse strategies. The remaining strategies with a moderate risk-aversion follow a compromise. The submitted bids in the day-ahead market, combined with the market-clearing results, determine the trader's position entering the intraday stage.

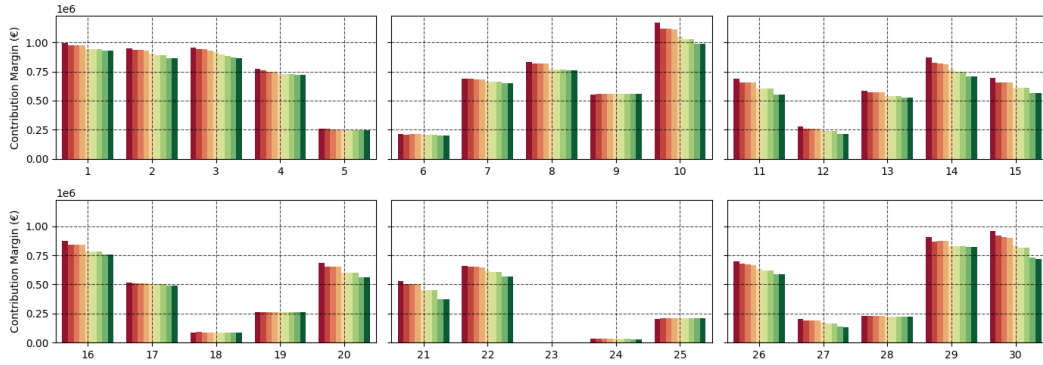


Figure 25: Day-ahead market contribution margins across different trading strategies and risk levels.

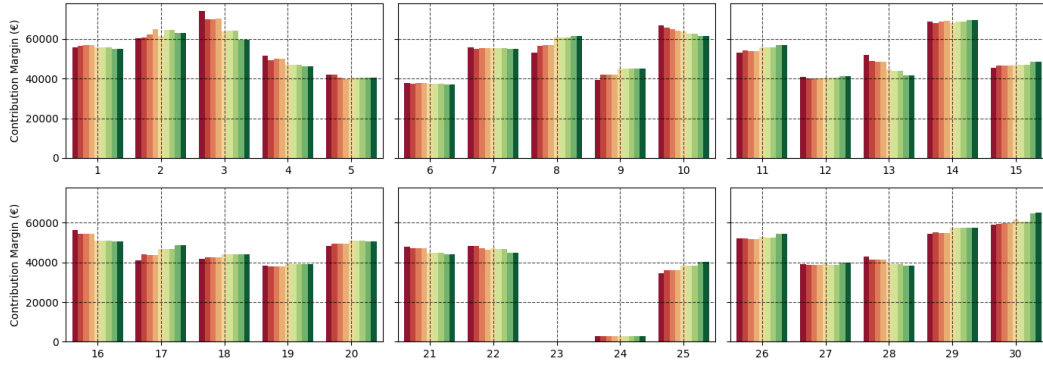


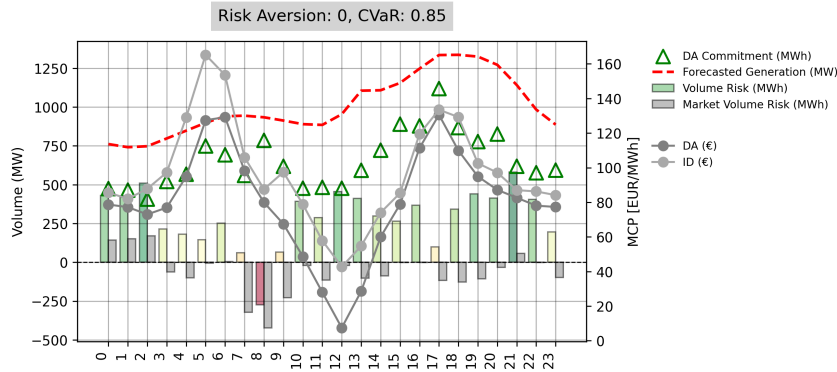
Figure 26: Intraday market contribution margins across different trading strategies and risk levels.

Unlike the day-ahead market, the intraday market serves as a corrective mechanism that allows traders to adjust their positions based on deviations from forecasted generation or unexpected market movements. In this setting, the risk-neutral strategy has the potential to generate higher profits when intraday market conditions are favorable. By delaying trading decisions until intraday sessions, the risk-neutral trader exploits price volatility to secure better deals. However, if intraday prices are unfavorable, the contribution margins of the risk-neutral strategy converge to those of the more conservative approaches, reducing the advantage of speculative

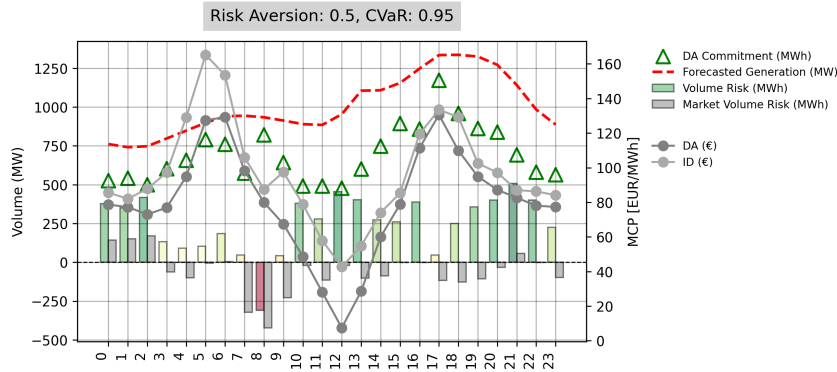
bidding.

Once the market-clearing price (MCP) is determined after the end of day-ahead auction, the final trading schedule is constructed by accumulating all accepted bids, i.e., those placed below the MCP for the corresponding hour of the day-ahead. This process determines the actual commitments for each trading strategy and provides insight into the volume risk, forecasted generation, scheduled commitments, and the overall impact of bidding decisions on market outcomes.

Figures 27 and 28 illustrate the differences in day-ahead commitment volumes and market exposure between the two strategies across two consecutive days. The risk-averse approach generally results in higher scheduled commitments in the day-ahead market due to its diversified bidding strategy and explicit incorporation of risk management measures. By distributing bids across a broader range of price levels, the risk-averse trader mitigates volume risk and maintains flexibility, ensuring that commitments are not concentrated at a single price point. This leads to a more balanced—though conservative—allocation of energy between the day-ahead and intraday markets.



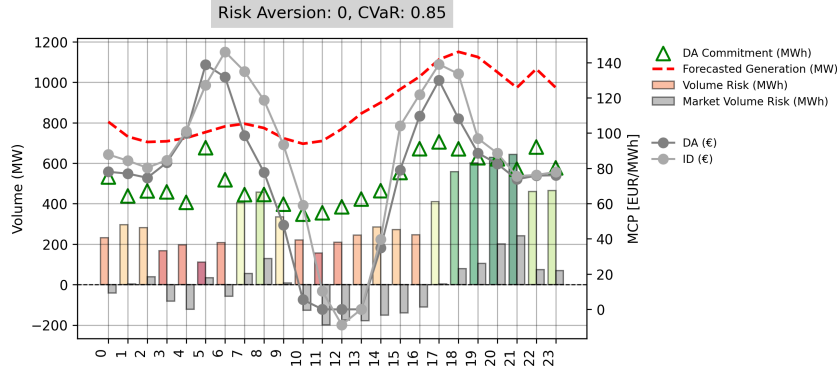
(a) Day-ahead market schedule for risk-neutral strategy ($\alpha = 0$, CVaR = 0.85).



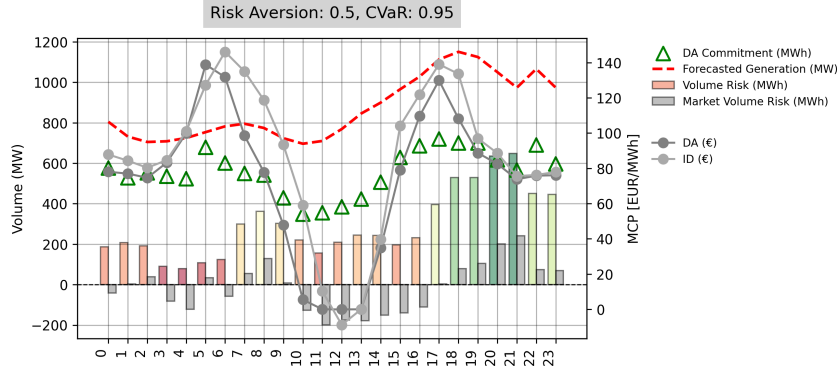
(b) Day-ahead market schedule for risk-averse strategy ($\alpha = 0.5$, CVaR = 0.95).

Figure 27: Comparison of scheduled commitments under risk-neutral and risk-averse strategies on 2024-09-17.

In both figures, it is important to note that 50% of the forecasted generation is pre-allocated—this threshold is visually represented as the midpoint of the red forecast generation line. Any bidding volumes above this midpoint reflect the optimizer’s strategic allocations, determined through the stochastic optimization model based on scenario probabilities, market expectations, and risk preferences.



(a) Day-ahead market schedule for risk-neutral strategy ($\alpha = 0$, $\text{CVaR} = 0.85$).



(b) Day-ahead market schedule for risk-averse strategy ($\alpha = 0.5$, $\text{CVaR} = 0.95$).

Figure 28: Comparison of scheduled commitments under risk-neutral and risk-averse strategies on 2024-09-18.

Further insights can be gained by analyzing the relationship between the day-ahead and the intraday ID3 Market Clearing Price (MCP) shown in the figures. For instance, in several hours on both September 17 and 18, the ID3 MCP is higher than the day-ahead MCP. The scenario tree effectively captures this price inversion, leading the risk-neutral strategy to strategically underbid in the day-ahead market to preserve flexibility. This behavior is clearly observable in Figures 27(a) and 28(a) show that several hourly commitments fall well below the forecasted generation line. When the forecasted generation turns out to be accurate—as it does during the early and late hours of both days—this strategy allows the trader to maximize expected contribution margins by shifting volume to the more favorable intraday market.

Conversely, the risk-averse strategy, shown in Figures 27(b) and 28(b), opts to commit volumes earlier—even when the intraday market might offer better prices. This reflects a deliberate prioritization of revenue stability over speculative profit, reducing exposure to price volatility and the risk of unaccepted bids. Notably, the commitment bars for the risk-averse strategy are more evenly distributed and consistently align closer to the forecasted generation, particularly in the morning and afternoon hours.

Additionally, the volume risk bars (below the x-axis) provide insight into the market exposure for each strategy. The risk-neutral model frequently leaves more forecasted volume uncommitted in the day-ahead market, especially when prices are expected to rise or when uncertainty is high. This increases exposure to real-time market conditions, which can be either an advantage or a vulnerability. In contrast, the risk-averse model reduces these uncommitted volumes significantly, signaling its preference for certainty and secured cash flows.

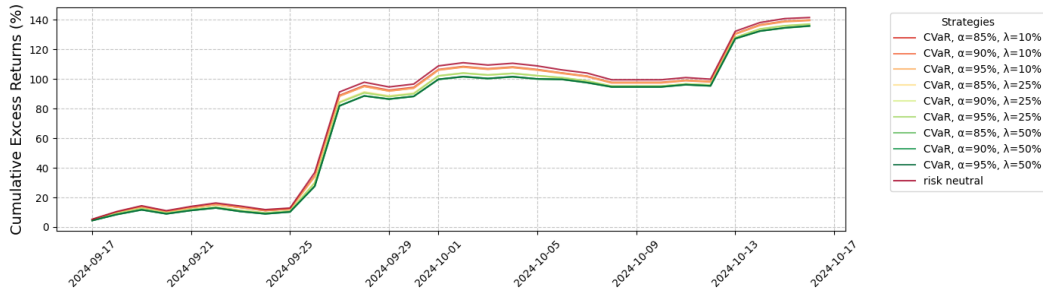
Regarding the problem size, solver parameters, and computational performance, we found the following configuration to be suitable. In its reduced form, the problem for a single day consists of approximately 2,741 variables, including 2,640 with quadratic terms, and around 300–800 constraints depending on the day. The model includes roughly 32,500 coefficients. It was implemented in Python and solved using the Gurobi 11.0.0 solver with a time limit of 300 seconds, a MIP gap of 0.01, and single-threaded execution due to license restrictions. The solver used the barrier method and typically required only 12–15 iterations, solving each daily instance in about 0.01 seconds. Presolve was highly effective, consistently reducing the number of constraints to around 48. Given the compact size and structure of the problem, even with quadratic terms, solution times remained negligible across all days.

7.1 Comparison with Vattenfall Benchmark

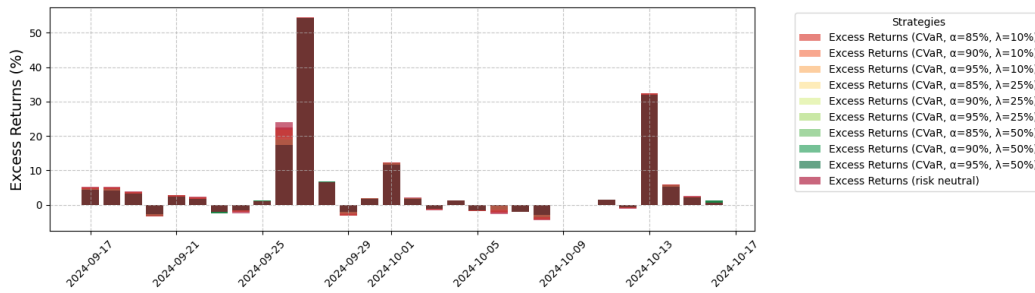
The benchmark strategy employed by Vattenfall follows a conservative, risk-averse approach, wherein 90% of the forecasted generation is sold in the day-ahead market to secure revenues upfront. This ensures a high degree of revenue certainty while limiting exposure to volatile intraday price fluctuations. Only 10% of the generation is retained for the intraday market, enabling limited flexibility to exploit favorable price movements.

In contrast, our optimized strategies adopt a dynamic, market-responsive allocation mechanism. Based on the generation schedule derived in Section 7, the day-ahead quantity is sold at the corresponding market clearing price (MCP).¹⁰ The actual realized generation, minus the day-ahead sold quantity, is then sold at the ID3 price in the intraday market, forming the basis of our total revenue. On the cost side, a uniform limit price per unit generation is applied, enabling a straightforward comparison of net profits across strategies.

Over a one-month testing period, the optimized strategies, particularly the risk-neutral configuration, consistently outperformed the benchmark. In terms of cumulative returns, the optimized risk-neutral strategy exceeded the benchmark by over 140% as shown in Figure 29a. This significant outperformance was achieved despite the excess returns exhibiting a stepwise pattern, where a few high-return days contributed disproportionately to overall gains. These spikes stemmed from high forecast accuracy and beneficial price dynamics, which the stochastic optimization framework effectively capitalized on.



(a) Line chart of cumulative excess returns.



(b) Bar chart of cumulative excess returns.

Figure 29: Excess returns compared to the Vattenfall benchmark across various risk strategies (September 17, 2024, to October 16, 2024).

Moreover, we observed in Figure 29b that risk-neutral strategies began outperforming

¹⁰ We assume that we are price takers in both the day-ahead and intraday markets, meaning all accepted bids are settled at the MCP and ID3 prices, respectively. This simplification implies no strategic market impact from individual bids.

risk-averse strategies during the later part of September. However, this trend reversed in early October due to a combination of worsening intraday price signals—including negative or highly unfavorable prices—and increased forecast errors. These factors led to a drop in profitability for the risk-neutral strategy, causing its returns to converge with the more conservative strategies.

Nonetheless, the performance remained robust overall, with consistent positive returns on most days. Periods of lower forecast generation—particularly between late September and October 10—did introduce challenges such as reduced bidding volumes and revenue potential. Despite this, the strategy maintained stable returns and effectively mitigated downside risk. These results highlight the capability of our approach to generate consistent excess returns under real-world constraints while dynamically managing risk exposure.

8 Conclusions and Outlook

This thesis presents a multi-stage stochastic optimization methodology that models the uncertainties associated with trading decisions across the day-ahead spot market and intraday spot market. Utilizing scenario trees and conditional expectations, the framework enables the derivation of optimal bidding strategies while accounting for risk preferences. The analysis highlights that risk-averse strategies favor securing revenues at earlier stages to minimize exposure to the volatility of the intraday market. Placing reserve bids at lower price levels and prioritizing sales in the day-ahead market effectively reduce risk, though at the potential cost of foregone higher revenues in the reserve and intraday markets. These insights are valuable for traders, policymakers, and researchers aiming to improve short-term market decision-making under uncertainty.

An important pillar of the proposed methodology is the generation of accurate scenario forecasts for prices and production. As emphasized throughout [Section 5.1.2](#) and [Section 5.2.1](#), reliable scenario trees are crucial because any systematic errors or biases in forecast modeling can propagate through the optimization process and significantly impact bidding outcomes. Particularly, residual diagnostics indicated that the baseline forecasting models could benefit from improvements in capturing serial correlation and volatility clustering. Incorporating advanced time series models, such as GARCH-type frameworks, could better address heteroscedasticity and non-constant

variance in residuals, thereby producing more realistic and statistically consistent scenarios. Improved scenario generation would not only enhance the quality of the optimization outputs but also increase the robustness of trading strategies against extreme market events.

Building on the developed methodology, several extensions can further enhance its applicability and accuracy. A key improvement is the inclusion of the imbalance market, which would provide a more comprehensive representation of real-time deviations and their financial impacts. Additionally, expanding the scenario generation process by incorporating diverse typedays and more sophisticated scenario reduction techniques would enhance the model's ability to capture a broader spectrum of market conditions. For real-world application, traders could integrate portfolio-specific forecasting data, including fuel and carbon prices, while transitioning from generic typedays to daily scenario trees to refine decision-making in live market environments.

Beyond trading strategies, the methodology can be extended to analyze market behavior during extreme events, such as energy scarcities, or in future electricity systems with high renewable penetration. Coupling this framework with other market models and insights from recent scarcity events would support policymakers in assessing regulatory and market design implications. Furthermore, integrating electricity storage solutions and sector coupling with alternative energy carriers, such as hydrogen, would introduce additional complexities that require advanced optimization techniques, particularly in reserve energy activation scenarios. Lastly, an extension toward investment appraisal under uncertain revenue streams would provide a structured approach for evaluating flexibility investments and optimizing multi-market participation.

In conclusion, this research contributes to the understanding of short-term electricity market dynamics under uncertainty, offering a robust framework for optimizing trading strategies. By incorporating imbalance markets, refining scenario tree methodologies, enhancing forecast models through volatility modeling, and extending the framework toward investment and policy analysis, the approach can be further strengthened to meet the evolving challenges of electricity markets. Future developments in this area will be instrumental in supporting traders and decision-makers in navigating an increasingly complex and uncertain energy landscape.

References

- [1] Benth, F. E., Kallsen, J., and Meyer-Brandis, T. (2007). A non-gaussian ornstein–uhlenbeck process for electricity spot price modeling and derivatives pricing. *Applied Mathematical Finance*, 14(2):153–169.
- [2] Birge, J. R. and Louveaux, F. (2011). *Introduction to Stochastic Programming*. Springer Series in Operations Research and Financial Engineering. Springer, New York, NY, 2 edition.
- [3] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- [4] Boomsma, T. K., Juul, N., and Fleten, S.-E. (2014). Bidding in sequential electricity markets: The nordic case. *European Journal of Operational Research*, 238(3):797–809.
- [5] Conejo, A. J., Carrion, M., and Morales, J. M. (2010). *Decision Making Under Uncertainty in Electricity Markets*. Springer.
- [6] Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236.
- [7] Deng, S.-J. and Oren, S. S. (2006). Electricity derivatives and risk management. *Energy*, 31(6–7):940–953.
- [8] Fleten, S.-E. and Kristoffersen, T. K. (2007). Stochastic programming for optimizing bidding strategies of a nordic hydropower producer. *European Journal of Operational Research*, 181(2):916–928.
- [9] Ho, N., Nguyen, X., Yurochkin, M., Bui, H. H., Huynh, V., and Phung, D. (2017). Multilevel clustering via wasserstein means. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1501–1509. PMLR.
- [10] Janczura, J., Trück, S., Weron, R., and Wolff, R. (2013). Identifying spikes and seasonal components in electricity spot price data: A guide to robust modeling. *Energy Economics*, 38:96–110.

- [11] Keles, D., Genoese, M., Möst, D., and Fichtner, W. (2012). Comparison of extended mean-reversion and time series models for electricity spot price simulation considering negative prices. *Energy Economics*, 34(4):1012–1032.
- [12] Kiesel, R. and Paraschiv, F. (2017). Econometric analysis of 15-minute intraday electricity prices. *Energy Economics*, 64:77–90.
- [13] Kremer, M., Kiesel, R., and Paraschiv, F. (2021). An econometric model for intraday electricity trading. *Philosophical Transactions of the Royal Society A*, 379(2202).
- [14] Laur, A., Nieto-Martin, J., Bunn, D. W., and Vicente-Pastor, A. (2020). Optimal procurement of flexibility services within electricity distribution networks. *European Journal of Operational Research*, 285(1):34–47.
- [15] Lv, S., Li, J., Guo, Y., and Shi, Z. (2019). A typical distributed generation scenario reduction method based on an improved clustering algorithm. *Applied Sciences*, 9(20):4262.
- [16] Meyer-Brandis, T. and Tankov, P. (2008). Multi-factor jump-diffusion models of electricity prices. *International Journal of Theoretical and Applied Finance*, 11(5):503–528.
- [17] Möst, D. and Keles, D. (2010). A survey of stochastic modelling approaches for liberalised electricity markets. *European Journal of Operational Research*, 207(2):543–556.
- [18] Ottesen, S., Tomasgard, A., and Fleten, S.-E. (2018). Multi-market bidding strategies for demand side flexibility aggregators in electricity markets. *Energy*, 149:120–134.
- [19] Russo, M. and Bertsch, V. (2020). A looming revolution: Implications of self-generation for the risk exposure of retailers. *Energy Economics*, 92:104970.
- [20] Russo, M., Kraft, E., Bertsch, V., and Keles, D. (2021). Short-term risk management for electricity retailers under rising shares of decentralized solar generation. Technical Report No. 57, NTNU, KIT, Ruhr-University Bochum, DTU.

- [21] Swider, D. J. (2007). Stochastic modeling for balancing power markets. *International Journal of Electrical Power & Energy Systems*, 29(7):565–578.
- [22] Uniejewski, B., Weron, R., and Ziel, F. (2018). Variance stabilizing transformations for electricity spot price forecasting. *IEEE Transactions on Power Systems*, 33(2):2219–2229.
- [23] Wozabal, D. and Rameseder, G. (2020). Optimal bidding of a virtual power plant on the spanish day-ahead and intraday market for electricity. *European Journal of Operational Research*, 280(2):639–655.
- [24] Zhao, S., Yao, J., and Li, Z. (2021). Wind power scenario reduction based on improved k-means clustering and sbr algorithm. *Power System Technology*, 45(10):3947–3954.
- [25] Zheng, Q. P., Wang, J., and Liu, A. L. (2015). Stochastic optimization for unit commitment—a review. *IEEE Transactions on Power Systems*, 30(4):1913–1924.
- [26] Ziel, F. (2020). The energy distance for ensemble and scenario reduction. *arXiv preprint arXiv:2005.14670*.

A Appendix

A.1 Stylized Facts of ID3 Prices

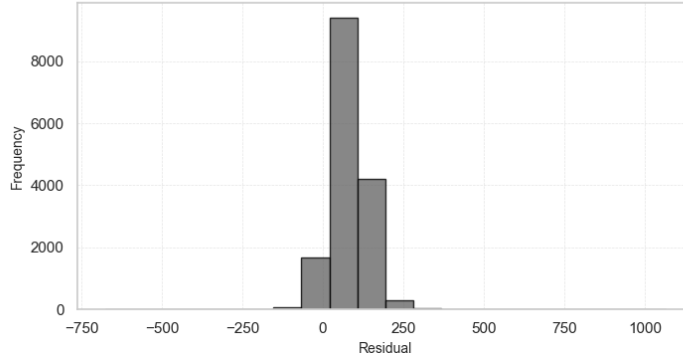


Figure 30: Histogram of ID3 prices showing distribution characteristics.



Figure 31: Autocorrelation function (ACF) and partial autocorrelation function (PACF) of intraday electricity prices.

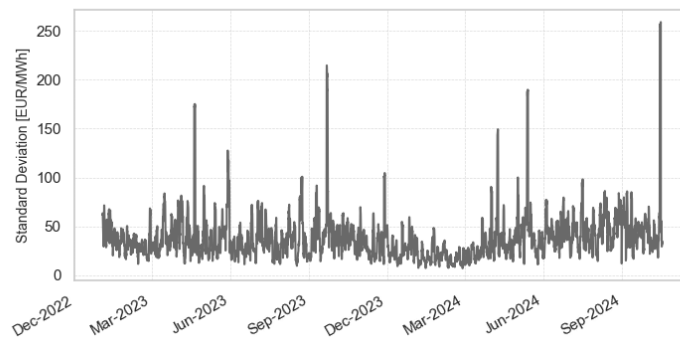


Figure 32: Volatility clustering observed in intraday price returns, measured as the rolling 30-period standard deviation (EUR/MWh).

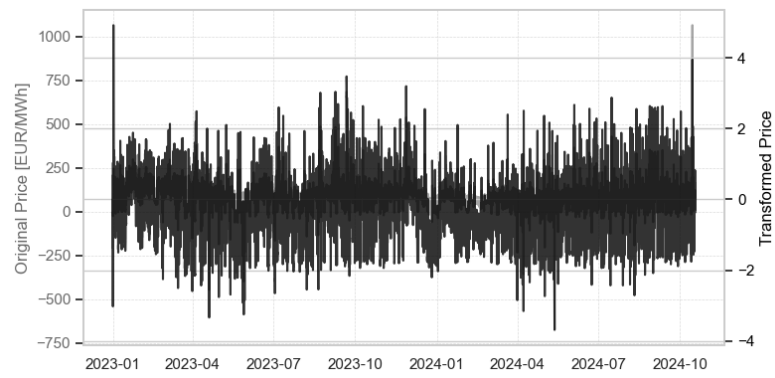


Figure 33: Original and arcsinh-Transformed intraday electricity prices.

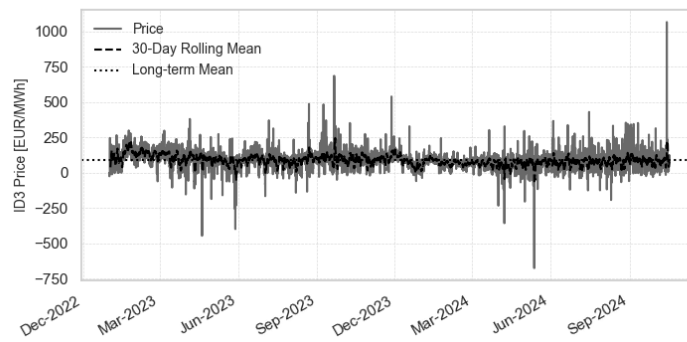


Figure 34: Evidence of mean reversion in ID3 prices over time.

A.2 Modeling of Market Segments

Table 7: Overview of the variables used in the empirical analysis

Variable (unit)	Frequency	Description	Data Source
Residual Load Forecasts (MW)	Hourly	Forecasted residual electricity demand based on total load and renewable production	Wattsight (Provider: EC12ENS)
ID3 price (EUR/MWh), ID	Hourly	Intraday electricity prices in the continuous trading	Vattenfall Energy Trading
Price (EUR/MWh), DA	Hourly	Market clearing price in the day-ahead auctions	Vattenfall Energy Trading
Forecasted Generation (MW)	Hourly	Forecasted electricity generation for the HKZ park in the Netherlands	Vattenfall Energy Trading
Actual Generation (MW)	Hourly	Actual electricity generation for the HKZ park in the Netherlands	Vattenfall Energy Trading

Table 8: OLS Regression Results for Day-Ahead Prices

Variable	Coefficient	Std. Error
Constant (β_0)	-13.5368***	1.325
Weekend (W_t)	-8.0079***	0.418
Price Min (P_{t-24}^{Min})	-0.0390***	0.005
Price Max (P_{t-24}^{Max})	-0.0448***	0.008
Last Price (P_{t-1}^{Last})	0.2612***	0.009
Lagged Price (P_{t-24})	0.3382***	0.010
Lagged Price (P_{t-48})	0.1650***	0.009
Lagged Price (P_{t-168})	0.2556***	0.008
Modular Price (M_t)	0.2320***	0.011
Residual Load Lag (RL_t)	0.0049***	6.57e-05
Residual Load Lag (RL_{t-24})	-0.0026***	7.86e-05
Residual Load Lag (RL_{t-48})	-0.0006***	7.36e-05
Residual Load Lag (RL_{t-168})	-0.0012***	7.02e-05
Model Statistics		
R-squared	0.784	
Adjusted R-squared	0.784	
Durbin-Watson	0.338	

*** indicates significance at the 5% level.

Table 9: OLS Regression Results for Intraday Prices

Variable	Coefficient	Std. Error
Constant (β_0)	-4.1257***	1.437
Day-Ahead Price (P_t^{DA})	0.8398***	0.009
Residual Load (RL_t)	0.0012***	0.000
Weekend (W_t)	-2.9785	1.934
Price Min (P_{t-24}^{Min})	-0.0339***	0.006
Price Max (P_{t-24}^{Max})	0.0156***	0.006
Last Price (P_{t-1}^{Last})	0.0956***	0.008
Lagged ID Price (P_{t-24}^{ID})	0.0345***	0.008
Lagged ID Price (P_{t-48}^{ID})	0.0134***	0.007
Lagged ID Price (P_{t-168}^{ID})	0.0169***	0.005
Modular Price (M_t)	0.0291***	0.014
Cycle * Weekend Interaction ($C_t \times W_t$)	0.0211	0.022
Residual Load Lag (RL_{t-1})	-0.0009***	0.000
Model Statistics		
R-squared	0.788	
Adjusted R-squared	0.788	
Durbin-Watson	0.545	

*** indicates significance at the 5% level.

A.3 Exemplary Scenario Tree

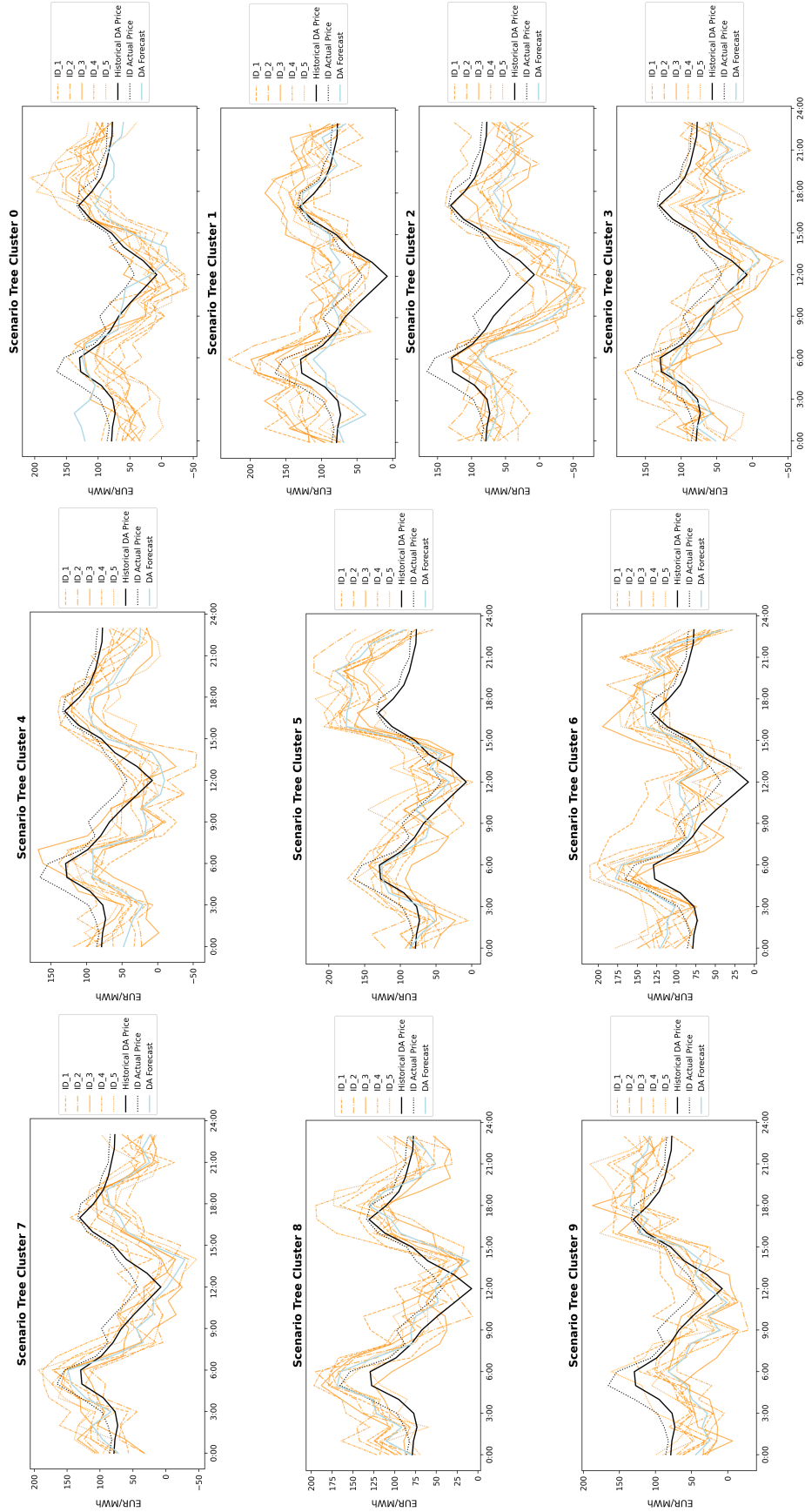


Figure 35: Visualization of Scenario Tree for September 17, 2024: Includes the day-ahead cluster mean forecast, original day-ahead prices, original intraday prices, and ten conditional intraday scenarios.

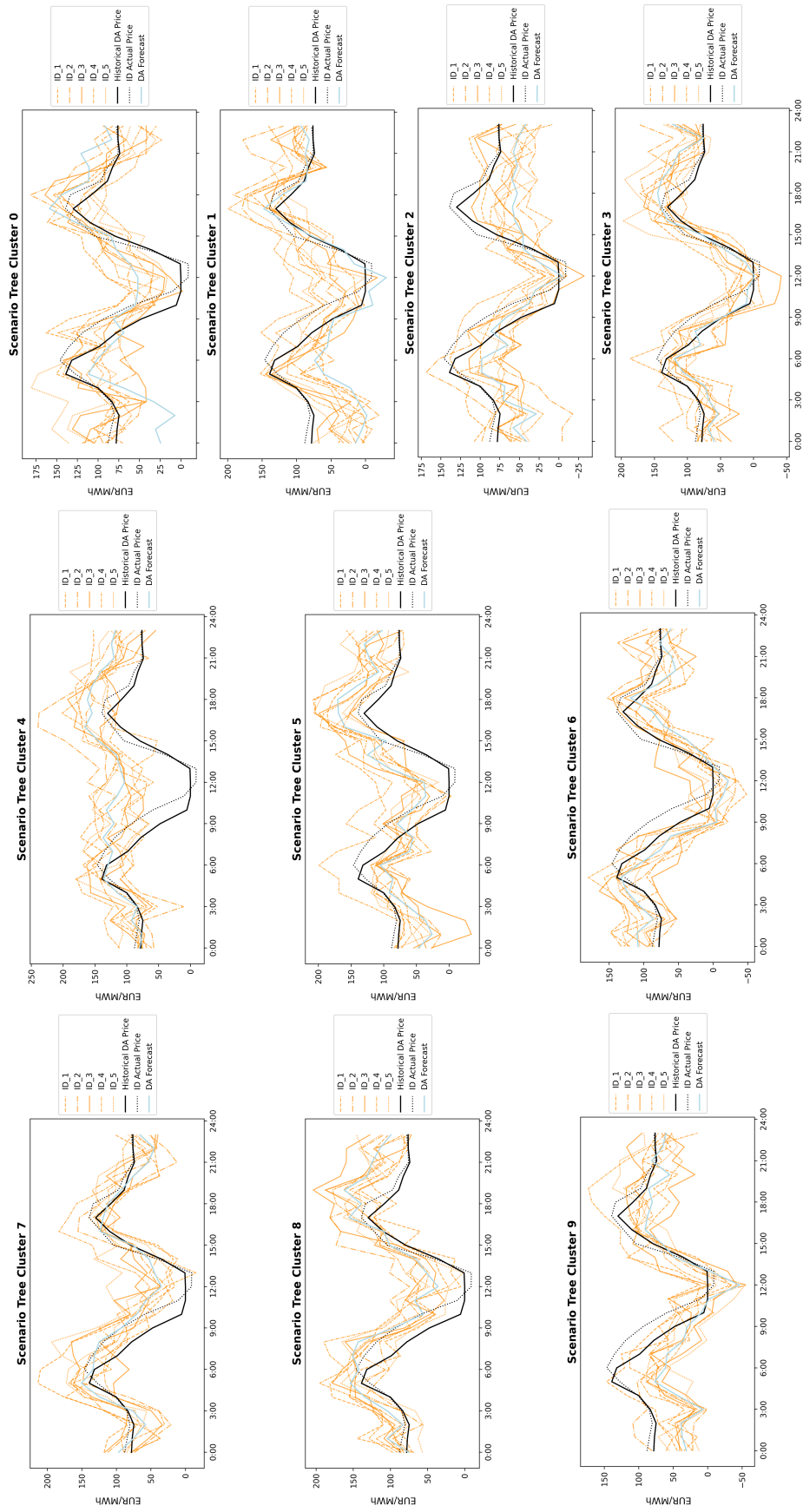


Figure 36: Visualization of Scenario Tree for September 18, 2024: Includes the day-ahead cluster mean forecast, original day-ahead prices, original intraday prices, and ten conditional intraday scenarios.

A.4 Definition and Modeling of Risk

The VaR η is defined as the $(1 - \alpha)$ -quantile of the contribution margin distribution.

For a discrete probability distribution, it is given by:

$$\text{VaR}(\alpha, x) = \max \{ \eta : \mathbb{P}_\omega [f(x, \omega) < \eta] \leq 1 - \alpha \}, \quad \forall \alpha \in (0, 1) \quad (30)$$

Here, f represents the distribution of contribution margins, x are deterministic parameters, and ω are stochastic parameters in the probability space Ω .

Including the VaR in the model requires modifying the objective function and adding auxiliary variables and constraints. The optimization objective becomes a weighted combination of the expected value and the VaR:

$$\max (1 - \lambda) \cdot \mathbb{E}_{(i,j,k) \in \Omega} (\pi_{i,j,k}) + \lambda \cdot \eta \quad (31)$$

with $\lambda \in (0, 1)$ as a parameter for risk aversion (e.g., $\lambda = 0.2$), and $\pi_{i,j,k}$ being the profit or contribution margin for scenario (i, j, k) .

The CVaR is defined as the expected value of the contribution margin in the worst $(1 - \alpha)$ outcomes, i.e., the expected loss when contribution margins fall below η .

For a discrete probability distribution:

$$\text{CVaR}(\alpha, x) = \max \left\{ \eta - \frac{1}{1 - \alpha} \cdot \mathbb{E}_{\omega \in \Omega} [\max\{\eta - f(x, \omega), 0\}] \right\} \quad (32)$$

The CVaR has the advantage of coherence and accounts for fat-tails in the distribution of contribution margins, offering a more robust risk assessment compared to VaR.

Table 10: Expected contribution margins and CVaR under different risk configurations

Risk metric	α [%]	λ [%]	$\mathbb{E}(\Pi)$ [EUR]	$\Delta\mathbb{E}(\Pi)$ [EUR]	CVaR [EUR]	ΔCVaR [EUR]
risk neutral	85	0	102251	-	-2506	-
CVaR	85	10	93156	9065	15506	18012
CVaR	85	25	80111	75503	26088	28594
CVaR	85	50	61236	41015	38010	40516
risk neutral	85	0	102251	-	-2506	-
CVaR	90	10	93102	9149	15601	18107
CVaR	90	25	80187	75552	26096	28599
CVaR	90	50	61231	40995	38054	40560
risk neutral	85	0	102251	-	-2506	-
CVaR	95	10	93099	9152	15608	18114
CVaR	95	25	80199	75553	26102	28608
CVaR	95	50	61158	41093	38105	40611

Affirmation

I hereby declare that I have composed my Master's thesis "*Stochastic Optimization of Trading Strategies in Sequential Electricity Markets*" independently using only those resources cited and that I have as such identified all passages which I have taken from publications verbatim or in substance. I agree that my thesis may be checked for plagiarism using testing software.

Neither this thesis nor any extract of it has been previously submitted to an examining authority in this or a similar form.

I have ensured that the written version of this thesis is identical to the version saved on the enclosed storage medium.

Kiel, 28.04.2025

Ahsan Muhammad